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United States Patent	12417430
Kind Code	B1
Date of Patent	September 16, 2025
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Verbally interactive materials handling facility

Abstract

Described are systems and techniques configured to acquire data from user speech generated by one user of a materials handling facility speaking to themselves or another user. The user speech is processed using natural language processing to determine key words, concepts, meanings, and so forth. Based on the processed data, queries may be executed, data may be presented in user interfaces, operation of the materials handling facility may be modified, and so forth.

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Appl. No.: **14/307053**

Filed: **June 17, 2014**

Publication Classification

Int. Cl.: **G06Q30/00** (20230101); **G06Q10/087** (20230101); **G10L25/48** (20130101); G10L15/065 (20130101)

U.S. Cl.:

CPC **G06Q10/087** (20130101); **G10L25/48** (20130101); G06Q30/00 (20130101); G10L15/065 (20130101)

Field of Classification Search

CPC: G10L (15/065)

USPC: 704/270

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Background/Summary

BACKGROUND
(1) Retailers, wholesalers, and other product distributors typically maintain an inventory of various items that may be ordered, purchased, leased, borrowed, rented, viewed, etc. by clients or customers. For example, an e-commerce website may maintain inventory in a fulfillment center. When a customer orders an item, the item is picked from inventory, routed to a packing station, packed and shipped to the customer. Likewise, physical stores maintain inventory in customer accessible areas (e.g., shopping area) and customers can pick items from inventory and take them to a cashier for purchase, rental, and so forth. Many of those physical stores also maintain inventory in a storage area, fulfillment center, or other facility that can be used to replenish inventory located in the shopping areas or to satisfy orders for items that are placed through other channels (e.g., e-commerce). Other examples of entities that maintain facilities holding inventory include libraries, museums, rental centers and the like. In each instance, for an item to be moved from one location to another, it is picked from its current location and transitioned to a new location. It is often desirable to provide to users information associated with the items in inventory, other users, or other information about operation of the facility.

Description

BRIEF DESCRIPTION OF FIGURES
(1) The detailed description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items or features.
(2) FIG. **1** is a block diagram illustrating a materials handling facility configured to respond to verbal communication between a user of the facility and another person, according to some implementations.
(3) FIG. **2** is a block diagram illustrating additional details of the materials handling facility, according to some implementations.
(4) FIG. **3** illustrates a block diagram of a server configured to support operation of the facility, according to some implementations.
(5) FIG. **4** is a block diagram of a tote, according to some implementations.
(6) FIG. **5** illustrates a user device configured to acquire information about user context, present information, and so forth, according to some implementations.
(7) FIG. **6** illustrates an overhead imaging sensor configured to acquire sensor data in the facility, according to some implementations.
(8) FIG. **7** illustrates a user interface for setting how the facility interacts with the user, according to some implementations.
(9) FIG. **8** illustrates user speech and the system responding to this speech, according to some implementations.
(10) FIG. **9** illustrates the system responding to speech between two users of the facility, according to some implementations.
(11) FIG. **10** depicts a flow diagram of a process for performing actions based on speech between users of the facility, according to some implementations.
(12) FIG. **11** depicts a flow diagram of another process for performing actions based on speech between users of the facility, according to some implementations.
(13) While implementations are described herein by way of example, those skilled in the art will recognize that the implementations are not limited to the examples or figures described. It should be understood that the figures and detailed description thereto are not intended to limit implementations to the particular form disclosed but, on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to.

DETAILED DESCRIPTION

(14) This disclosure describes systems and techniques for providing information to a user of a materials handling facility (facility). The facility may include, or have access to, an inventory management system. The inventory management system may be configured to maintain information about items, users, condition of the facility, and so forth. For example, the inventory management system may maintain data indicative of what items a particular user is ordered to pick, historical data about items the particular user has picked, suggested routes within the facility to follow while picking, and so forth.

(15) The systems and techniques described in this disclosure may be used to provide a user of the facility with information, modify operation of the facility or mechanisms therein, or otherwise assist the user. A plurality of imaging sensors configured to acquire image data, such as cameras, may also be arranged within the facility. The image data may be processed by the inventory management system to determine information such as identity of a user, location of the user, and so forth.

(16) A plurality of microphones may be arranged within the facility and configured to acquire audio data representative of sound within the facility. The microphones may be fixed within the facility, mounted on totes, carried by the user, and so forth. In some implementations, a plurality of microphones may be configured in an array. Data acquired by the array may be processed using techniques such as beamforming to allow for localization of a sound. Localization of a sound includes determining a position in space from which the sound emanates. The data acquired by the microphones may be processed by a speech interaction system of the inventory management system.

(17) The speech interaction system may be configured to process the audio data and determine a communicative sound issued from the user. The communicative sounds may include, but are not limited to, speech, non-lexical utterances such as laughs, sighs, snorts, fingersnaps, handclaps, and so forth. For ease of discussion and not necessarily by way of limitation, speech as used in this disclosure may include the communicative sounds described above. The speech interaction system may be configured to recognize speech to generate text or commands, voice recognition to identify a particular user based on the communicative sounds, and so forth. In some implementations the speech interaction system may use the image data or data from other sensors to localize a source of sound. For example, the speech interaction system may compare facial movements with sounds to determine a particular user is speaking.

(18) The speech interaction system is configured to process speech which is not directed to the speech interaction system. For example, the speech interaction system may process speech while a user is talking to themselves, speech when the user is talking to another person, speech when the user is talking to another system, and so forth. The other person or system to whom the user speaking may be physically present in the facility, may be in communication with the user by way of a cell phone or other audible communication mechanism, and so forth. For example, the user may be speaking to a personal information assistant application which executes at least in part on servers outside of the facility. The speech which is processed by the speech interaction system may not necessarily have a predefined syntax or grammar, instead comprising natural language. The speech interaction system may analyze attributes such as intonation, speaking rate, perplexity, and so forth to determine when speech is or is not directed to the speech interaction system. For example, based on an intonation which indicates a particular pattern of pitch, a speaking rate above 150 words per minute, and perplexity above a threshold value, the speech may be deemed to be directed at another user.

(19) The speech interaction system determines one or more actions to perform based on the speech. These actions may include one or more of: retrieving information, storing information, generating a user interface for presentation to the user by way of an output device such as a speaker, moving inventory locations within the facility, moving inventory from one inventory location to another, requesting assistance from another user, and so forth. For example, the speech interaction system may hear the user say to another user “I can't find it” and determine that the user is unable to find a particular item to pick. The speech interaction system may then perform the actions of retrieving information indicating an inventory location of the particular item, generating user interface data including synthesized speech which is then presented using a speaker for the user to hear as an audible user interface. Continuing the example, synthesized speech may announce “The widget on your pick list is located ahead of you on rack A shelf 3.” Thus, the user is presented with information which may be useful in performing their tasks in the facility, without having to initiate a query with the inventory management system.

(20) The determination of the one or more actions to perform by the speech interaction system may be based at least in part on a user context. The user context provides information which may provide cues to the speech interaction system to help disambiguate words, determine meaning, select particular actions, and so forth. For example, the user context may include a direction of gaze of the user, proximity to other users, location of the user within the facility, and so forth. The user context may also include information which is specific to the user. For example, the user context may include historical activity such as a list of items picked since entry to the facility today.

(21) The inventory management system may be configured to identify the user. Identification may comprise associating a user with a particular user account. For example, the identification may involve one or more of facial recognition, entry of one or more login credentials such as the username and password, detection of a tag receiving data stored within the tag such as a radiofrequency identification (RFID) chip, and so forth. Once identified, the user context may also include user data which is specific to a particular identity. For example, the user data may include demographic data such as the user's age, relationship data indicative of a relationship between the identified user and another person, user preferences indicative of a preferred language, preferred wordbase, and so forth. Continuing the example, when the user's preferred language is English and the preferred wordbase is British English, audio data of the phrase “where is the torch” as added by the user may be correctly interpreted as an inquiry to find a flashlight item, and information indicating the location of the flashlight item may be presented in a user interface. In another example, the relationship data may provide additional meaning or context within which an appropriate action may be selected. For example, relationship data indicating that the user is a supervisor of a second user may result in interpretation of the phrase “could you please pick the widget from our overstock area” as an order to change the pick location for the item to the inventory location in the overstock area. In contrast, where the relationship data indicates that the user is a subordinate of the second user, the action may comprise presenting a user interface to the second user which includes a prompt to confirm the change in pick location.

(22) The facility may include a materials handling facility, library, museum, and so forth. As used herein, a materials handling facility may include, but is not limited to, warehouses, distribution centers, cross-docking facilities, order fulfillment facilities, packaging facilities, shipping facilities, rental facilities, libraries, retail stores, wholesale stores, museums, or other facilities or combinations of facilities for performing one or more functions of materials (inventory) handling.

(23) The systems and techniques described herein enable actions to be performed in the facility such as information presentation or movement of inventory in a way which is meaningful, unobtrusive, and responsive to the user without the user specifically requesting those actions. The speech interaction system may improve overall operation of the inventory management system, improve the experience of the users, and so forth.

(24) Illustrative System

(25) An implementation of a materials handling system **100** configured to store and manage inventory items is illustrated in FIG. **1**. A materials handling facility **102** (facility) comprises one or more physical structures or areas within which one or more items **104**(**1**), **104**(**2**), . . . , **104**(Q) may be held. As used in this disclosure, letters in parenthesis such as “(Q)” indicate an integer value. The items **104** comprise physical goods, such as books, pharmaceuticals, repair parts, electronic gear, and so forth.

(26) The facility **102** may include one or more areas designated for different functions with regard to inventory handling. In this illustration, the facility **102** includes a receiving area **106**, a storage area **108**, and a transition area **110**.

(27) The receiving area **106** may be configured to accept items **104**, such as from suppliers, for intake into the facility **102**. For example, the receiving area **106** may include a loading dock at which trucks or other freight conveyances unload the items **104**.

(28) The storage area **108** is configured to store the items **104**. The storage area **108** may be arranged in various physical configurations. In one

implementation, the storage area **108** may include one or more aisles **112**. The aisle **112** may be configured with, or defined by, inventory locations **114** on one or both sides of the aisle **112**. The inventory locations **114** may include one or more of shelves, racks, cases, cabinets, bins, floor locations, or other suitable storage mechanisms. The inventory locations **114** may be affixed to the floor or another portion of the facility's structure, or may be movable such that the arrangements of aisles **112** may be reconfigurable. In some implementations, the inventory locations **114** may be configured to move independently of an outside operator. For example, the inventory locations **114** may comprise a rack with a power source and a motor, operable by a computing device to allow the rack to move from one position within the facility **102** to another. Continuing the example, the inventory location **114** may move from one aisle **112** to another, from one position within an aisle **112** to another, and so forth.
(29) One or more users **116** and totes **118** or other material handling apparatus may move within the facility **102**. For example, the user **116** may move about within the facility **102** to pick or place the items **104** in various inventory locations **114**, placing them on the tote **118** for ease of transport.
(30) Instead of or in addition to the users **116** other mechanisms such as robots, forklifts, cranes, aerial drones, conveyors, elevators, pipes, and so forth, may move items **104** about the facility **102**. For example, a robot may pick the item **104** from a first inventory location **114**(**1**) and move it to a second inventory location **114**(**2**).
(31) The one or more users **116** may play different roles in the operation of the facility **102**. For example, some users **116** may be tasked with picking or placing items **104** while other users **116** support the operation of the facility **102** by assisting other users **116**. Furthermore, the roles of a particular user **116** may change over time. For example the user **116**(**1**) may start out as an assistant and later be tasked as a supervisor.
(32) Different relationships may exist between users **116**. For example the user **116**(**1**) may be a supervisor of the user **116**(**2**). In another example, the user **116**(**1**) may be a sibling, spouse, friend, or otherwise be related to the user **116**(**2**).
(33) One or more sensors **120** may be configured to acquire information in the facility **102**. The sensors **120** may include, but are not limited to, microphones, imaging sensors, weight sensors, radio frequency (RF) receivers, temperature sensors, humidity sensors, vibration sensors, and so forth. The sensors **120** may be stationary or mobile, relative to the facility **102**. For example, the inventory locations **114**, the totes **118**, or other devices such as user devices may contain microphones configured to acquire audio data. In another example, the inventory locations **114** may contain imaging sensors configured to acquire images of pick or placement of items **104** on shelves. The sensors **120** are discussed in more detail below with regard to FIG. **2**.
(34) During operation of the facility **102**, the sensors **120** may be configured to provide information suitable for tracking how objects move within the facility **102**. For example, a series of images acquired by an imaging sensor may indicate removal of an item **104** from a particular inventory location **114** by the user **116** and placement of the item **104** on or at least partially within the tote **118**. The tote **118** is discussed in more detail below with regard to FIG. **4**.
(35) While the storage area **108** is depicted as having one or more aisles **112**, inventory locations **114** storing the items **104**, sensors **120**, and so forth, it is understood that the receiving area **106**, the transition area **110**, or other areas of the facility **102** may be similarly equipped. Furthermore, the arrangement of the various areas within the facility **102** are depicted functionally rather than schematically. For example, in some implementations multiple different receiving areas **106**, storage areas **108**, and transition areas **110** may be interspersed rather than segregated.
(36) The facility **102** may include, or be coupled to, an inventory management system **122**. The inventory management system **122** is configured to interact with users **116** or devices such as sensors **120**, robots, material handling equipment, computing devices, and so forth, in one or more of the receiving area **106**, the storage area **108**, or the transition area **110**.
(37) The facility **102** may be configured to receive different kinds of items **104** from various suppliers, and to store them until a customer orders or retrieves one or more of the items **104**. A general flow of items **104** through the facility **102** is indicated by the arrows of FIG. **1**. Specifically, as illustrated in this example, items **104** may be received from one or more suppliers, such as manufacturers, distributors, wholesalers, and so forth, at the receiving area **106**. In various implementations, the items **104** may include merchandise, commodities, perishables, or any suitable type of item, depending on the nature of the enterprise that operates the facility **102**.
(38) Upon being received from a supplier at receiving area **106**, the items **104** may be prepared for storage. For example, in some implementations, items **104** may be unpacked or otherwise rearranged. The inventory management system **122** may include one or more software applications executing on a computer system to provide inventory management functions. These inventory management functions may include maintaining information indicative of the type, quantity, condition, cost, location, weight, or any other suitable parameters with respect to the items **104**. The items **104** may be stocked, managed, or dispensed in terms of countable, individual units or multiples, such as packages, cartons, crates, pallets, or other suitable aggregations. Alternatively, some items **104**, such as bulk products, commodities, and so forth, may be stored in continuous or arbitrarily divisible amounts that may not be inherently organized into countable units. Such items **104** may be managed in terms of measurable quantity such as units of length, area, volume, weight, time, duration, or other dimensional properties characterized by units of measurement. Generally speaking, a quantity of an item **104** may refer to either a countable number of individual or aggregate units of an item **104** or a measurable amount of an item **104**, as appropriate.
(39) After arriving through the receiving area **106**, items **104** may be stored within the storage area **108**. In some implementations, like items **104** may be stored or displayed together in the inventory locations **114** such as in bins, on shelves, hanging from pegboards, and so forth. In this implementation, all items **104** of a given kind are stored in one inventory location **114**. In other implementations, like items **104** may be stored in different inventory locations **114**. For example, to optimize retrieval of certain items **104** having frequent turnover within a large physical facility, those items **104** may be stored in several different inventory locations **114** to reduce congestion that might occur at a single inventory location **114**.
(40) When a customer order specifying one or more items **104** is received, or as a user **116** progresses through the facility **102**, the corresponding items **104** may be selected or “picked” from the inventory locations **114** containing those items **104**. In various implementations, item picking may range from manual to completely automated picking. For example, in one implementation, a user **116** may have a list of items **104** they desire and may progress through the facility **102** picking items **104** from inventory locations **114** within the storage area **108**, and placing those items **104** into a tote **118**. In other implementations, employees of the facility **102** may pick items **104** using written or electronic pick lists derived from customer orders. These picked items **104** may be placed into the tote **118** as the employee progresses through the facility **102**.
(41) After items **104** have been picked, they may be processed at a transition area **110**. The transition area **110** may be any designated area within the facility **102** where items **104** are transitioned from one location to another, or from one entity to another. For example, the transition area **110** may be a packing station within the facility **102**. When the item **104** arrives at the transition area **110**, the item **104** may be transitioned from the storage area **108** to the packing station. Information about the transition may be maintained by the inventory management system **122**.
(42) In another example, if the items **104** are departing the facility **102** a list of the items **104** may be obtained and used by the inventory management system **122** to transition responsibility for, or custody of, the items **104** from the facility **102** to another entity. For example, a carrier may accept the items **104** for transport with that carrier accepting responsibility for the items **104** indicated in the list. In another example, a customer may purchase or rent the items **104** and remove the items **104** from the facility **102**.
(43) The user **116** may produce communicative sounds while at the facility **102**. For ease of illustration and not necessarily by way of limitation the communicative sounds are referred to herein as “speech” and may include, but are not limited to, audible verbalizations, non-lexical utterances, and so forth. The non-lexical utterances may include sighs, gasps, grunts, clicks, pops, or other sounds generated by the user **116**. Other communicative sounds may include those generated by an action of the user **116**, such as fingersnaps, handclaps, foot stomps, and so forth.
(44) The inventory management system **122** may include a speech interaction system **124** configured to access user context data **126** which is indicative of a user context to process user speech **128** in the audio data, and perform one or more actions. In some implementations, the one or more

actions may include selecting one or more output devices **130** and processing user interface data to produce a user interface **132** for presentation to the user **116**. The speech interaction system **124** may include a speech recognition module, voice recognition module, a natural language understanding (NLU) module, a semantic analysis module, or other modules. The speech recognition module is configured to process audio of the user speech **128** and produce text or commands representative of the speech. The voice recognition module is configured to distinguish one user **116** from another user **116** based on the sounds emitted by the user **116**. The NLU module is configured to parse the information generated by the modules, such as from the speech recognition module, to generate information indicative of meaning of the user speech **128**. The NLU module may use Hidden Markov Models (HMMs), neural networks, or other techniques to generate speech from the audio. For example, neural networks may be used to pre-process the audio data and perform functions such as feature transformation, dimensionality reduction, and so forth, with the resulting output used for HMM-based recognition.

(45) The user context data **126** may include information about one or more of: operation of the facility **102**, data about the items **104**, the user **116** as an anonymous individual, the user **116** as an identified individual, data external to the facility **102** such as weather or delivery route information, and so forth. For example, the user context data **126** may include position data indicative of a location of the one or more users **116** within the facility **102**. In another example, of the user context data **126** may comprise user data such as user preferences which have been previously acquired and stored.

(46) The user context data **126** may include data indicative of one or more items **104** within the facility **102** that the user **116** is interacting with. The interaction may include direct physical manipulation, such as picking up or touching the item **104**. The interaction may include looking at the item **104**, pointing, or gesturing at the item **104**. The interaction may include items **104** which are within a predetermine distance of the user **116**, such as within 2 meters.

(47) In some implementations the user context data **126** may include information from other users **116** or groups of users **116** who have been determined to be similar in one or more respects. For example, the user context data **126** may include information such as user preferences as set by other users **116** with similar demographic data, who have similar access permissions, who were in about the same location within the facility **102**, and so forth. This information may be used in place of information which is not yet available about the user **116**, or may supplement information associated with the user **116**.

(48) The user context data **126** may be used by the speech interaction system **124** to facilitate the functions thereof, such as recognizing words and determining meaning of the user speech **128**. The user context data **126** provides information which may provide cues to help disambiguate words, determine meaning, select particular actions, and so forth. For example, the user context data **126** may include a direction of a gaze of the user **116** which is indicative of a particular item **104** the user **116** is looking at. Using this information, the speech interaction system **124** may disambiguate the user speech **128** of “what is that” to determine that the user **116** is uncertain as to some attribute of the item **104**. For example, the user context may include a direction of a gaze of the user **116**, proximity to other users **116**, location of the user **116** within the facility **102**, and so forth. The user context may also include information which is specific to the user **116**. For example, the user context may include historical activity such as a list of items picked since entry to the facility **102** today.

(49) The user context data **126** may be used by the speech interaction system **124** to modify statistical models, assign weights in neural networks, select speech recognition lexicons indicative of how words are pronounced, select speech recognition grammars indicative of an expected form or structure of spoken words, and so forth. In implementations where the speech interaction system **124** uses HMM-based recognition techniques, the user context data **126** may be used to configure one or more of the HMMs in the module. For example, the user context data **126** may be used to modify weightings in context dependency for phonemes, to assign weightings to possible transcriptions of the speech, and so forth. Continuing the example, the user context data **126** may be used to assess which ones of a set of possible transcriptions should be retained. Thus, where the user context data **126** indicates the user **116** has a gaze directed at a product in the category “ice cream”, the transcription of “ice cream” may be retained while transcriptions “I scream”, “eye screem”, “ice ream”, and “high screen” are disregarded.

(50) A user device **134** may also be present within the facility **102**. The user device **134** may be provided by the user **116**, or may be provided by an operator of the facility **102** or another entity for use by the user **116**. The user device **134** may comprise a tablet computer, smartphone, notebook computer, data access terminal, wearable computing device, and so forth. The user device **134** may be used to acquire audio data, process user interface data to present the user interface **132**, or provide other functions. For example, information from the user device **134** such as message data, call metadata, application state data, and so forth may be included in the user context data **126**.

(51) The user **116** may benefit from various actions taken by the inventory management system **122** responsive to user speech **128**. For example, as the user **116**(**1**) moves about the facility **102** speaking to another person such as the user **116**(**2**) or another system, the speech interaction system **124** may process the user speech **128** and determine actions such as presenting information as output, moving inventory locations **114** or items **104**, and so forth. Instead of the user **116** explicitly querying inventory management system **122**, the system may instead determine and perform the actions unobtrusively. As a result, the experience of the user **116** in the facility **102** may be improved, efficiency of the facility's **102** operation may be improved, or other benefits may accrue.

(52) FIG. **2** is a block diagram **200** illustrating additional details of the facility **102**, according to some implementations. The facility **102** may be connected to one or more networks **202**, which in turn connect to one or more servers **204**. The network **202** may include private networks, public networks such as the Internet, or a combination thereof. The network **202** may utilize wired technologies (e.g., wires, fiber optic cable, and so forth), wireless technologies (e.g., radio frequency, infrared, acoustic, optical, and so forth), or other connection technologies. The network **202** is representative of any type of communication network, including one or more of data networks or voice networks. The network **202** may be implemented using wired infrastructure (e.g., copper cable, fiber optic cable, and so forth), a wireless infrastructure (e.g., cellular, microwave, satellite), or other connection technologies.

(53) The servers **204** may be configured to execute one or more modules or software applications associated with the inventory management system **122**. While the servers **204** are illustrated as being in a location outside of the facility **102**, in other implementations at least a portion of the servers **204** may be located at the facility **102**. The servers **204** are discussed in more detail below with regard to FIG. **3**.

(54) The users **116**, the totes **118**, or other objects in the facility **102** may be equipped with one or more radio frequency (RF) tags **206**. The RF tags **206** are configured to emit an RF signal **208**. In one implementation, the RF tag **206** may be a radio frequency identification (RFID) tag configured to emit the RF signal **208** upon activation by an external signal. For example, the external signal may comprise a radio frequency signal or a magnetic field configured to energize or activate the RFID tag. In another implementation, the RF tag **206** may comprise a transmitter and a power source configured to power the transmitter. For example, the RF tag **206** may comprise a Bluetooth Low Energy (BLE) transmitter and battery. In other implementations, the tag may use other techniques to indicate presence. For example, an acoustic tag may be configured to generate an ultrasonic signal which is detected by corresponding acoustic receivers. In yet another implementation, the tag may be configured to emit an optical signal.

(55) The inventory management system **122** may be configured to use the RF tags **206** for one or more of identification of the object, determining a position of the object, and so forth. For example, the users **116** may wear RF tags **206**, the totes **118** may have RF tags **206** affixed, and so forth which may be read and, based at least in part on signal strength, used to determine identity and position. The tote **118** is configured to carry or otherwise transport one or more items **104**. For example, the tote **118** may include a basket, a cart, a bag, and so forth. The tote **118** is discussed in more detail below with regard to FIG. **4**.

(56) Generally, the inventory management system **122** or other systems associated with the facility **102** may include any number and combination of input components, output components, and servers **204**.

(57) The one or more sensors **120** may be arranged at one or more locations within the facility **102**. For example, the sensors **120** may be mounted on or within a floor, wall, or ceiling, at an inventory location **114**, on the tote **118**, in the user device **134**, may be carried or worn by the user **116**, and so forth.
(58) The sensors **120** may include one or more imaging sensors **120**(**1**). These imaging sensors **120**(**1**) may include cameras configured to acquire images of a scene. The imaging sensors **120**(**1**) are configured to detect light in one or more wavelengths including, but not limited to, terahertz, infrared, visible, ultraviolet, and so forth. The inventory management system **122** may use image data acquired by the imaging sensors **120**(**1**) during operation of the facility **102**. For example, the inventory management system **122** may identify items **104**, users **116**, totes **118**, and so forth based at least in part on their appearance within the image data.
(59) One or more 3D sensors **120**(**2**) may also be included in the sensors **120**. The 3D sensors **120**(**2**) are configured to acquire spatial or three-dimensional data, such as depth information, about objects within a sensor field-of-view. The 3D sensors **120**(**2**) include range cameras, lidar systems, sonar systems, radar systems, structured light systems, stereo vision systems, optical interferometry systems, and so forth. The inventory management system **122** may use the three-dimensional data acquired to identify objects, determine a position of an object, and so forth. For example, the user context data **126** may include a position of the user **116** in three-dimensional space within the facility **102**.
(60) One or more buttons **120**(**3**) may be configured to accept input from the user **116**. The buttons **120**(**3**) may comprise mechanical, capacitive, optical, or other mechanisms. For example, the buttons **120**(**3**) may comprise mechanical switches configured to accept an applied force from a touch of the user **116** to generate an input signal. The inventory management system **122** may use data from the buttons **120**(**3**) to receive information from the user **116**. For example, the tote **118** may be configured with a button **120**(**3**) configured such that the button **120**(**3**) may be activated by the user **116**.
(61) The sensors **120** may include one or more touch sensors **120**(**4**). The touch sensors **120**(**4**) may use resistive, capacitive, surface capacitance, projected capacitance, mutual capacitance, optical, Interpolating Force-Sensitive Resistance (IFSR), or other mechanisms to determine the position of a touch or near-touch. For example, the IFSR may comprise a material configured to change electrical resistance responsive to an applied force. The position of that change in electrical resistance within the material may indicate the position of the touch. The inventory management system **122** may use data from the touch sensors **120**(**4**) to receive information from the user **116**. For example, the touch sensor **120**(**4**) may be integrated with the tote **118** to provide a touchscreen with which the user **116** may select from a menu one or more particular items **104** for picking.
(62) One or more microphones **120**(**5**) may be configured to acquire audio data indicative of sound present in the environment. The sound may include the user speech **128** uttered by the user **116.** In some implementations, arrays of microphones **120**(**5**) may be used. These arrays may implement beamforming techniques to provide for directionality of gain. The speech interaction system **124** may use the one or more microphones **120**(**5**) to accept voice input from the users **116**, determine the position of one or more users **116** in the facility **102**, and so forth. For example, the microphones **120**(**5**) may use time difference of arrival (TDOA) techniques to localize a position of a sound.
(63) One or more weight sensors **120**(**6**) are configured to measure the weight of a load, such as the item **104**, the user **116**, the tote **118**, and so forth. The weight sensors **120**(**6**) may be configured to measure the weight of the load at one or more of the inventory locations **114**, the tote **118**, or on the floor of the facility **102**. The weight sensors **120**(**6**) may include one or more sensing mechanisms to determine weight of a load. These sensing mechanisms may include piezoresistive devices, piezoelectric devices, capacitive devices, electromagnetic devices, optical devices, potentiometric devices, microelectromechanical devices, and so forth. The sensing mechanisms may operate as transducers which generate one or more signals based on an applied force, such as that of the load due to gravity. The inventory management system **122** may use the data acquired by the weight sensors **120**(**6**) to identify an object, determine a location of an object, maintain shipping records, and so forth.
(64) The sensors **120** may include one or more light sensors **120**(**7**). The light sensors **120**(**7**) may be configured to provide information associated with ambient lighting conditions such as a level of illumination. Information acquired by the light sensors **120**(**7**) may be used by the inventory management system **122** to adjust a level, intensity, or configuration of the output device **130**.
(65) One more radio frequency identification (RFID) readers **120**(**8**), near field communication (NFC) systems, and so forth may also be provided in the sensors **120**. For example the RFID readers **120**(**8**) may be configured to read the RF tags **206**. Information acquired by the RFID reader **120**(**8**) may be used by the inventory management system **122** to identify an object associated with the RF tag **206** such as the item **104**, the user **116**, the tote **118**, and so forth.
(66) One or more RF receivers **120**(**9**) may also be provided. In some implementations the RF receivers **120**(**9**) may be part of transceiver assemblies. The RF receivers **120**(**9**) may be configured to acquire RF signals **208** associated with Wi-Fi, Bluetooth, ZigBee, 3G, 4G, LTE, or other wireless data transmission technologies. The RF receivers **120**(**9**) may provide information associated with data transmitted via radio frequencies, signal strength of RF signals **208**, and so forth. For example, information from the RF receivers **120**(**9**) may be used by the inventory management system **122** to determine a location of an RF source such as the user device **134**.
(67) The sensors **120** may include one or more accelerometers **120**(**10**), which may be worn or carried by the user **116**, mounted to the tote **118**, and so forth. The accelerometers **120**(**10**) may provide information such as the direction and magnitude of an imposed acceleration. Data such as rate of acceleration, determination of changes in direction, speed, and so forth may be determined using the accelerometers **120**(**10**).
(68) A gyroscope **120**(**11**) provides information indicative of rotation of an object affixed thereto. For example, the tote **118**, the user device **134**, or other objects may be equipped with a gyroscope **120**(**11**) to provide user context data **126** indicative of a change in orientation.
(69) A magnetometer **120**(**12**) may be used to determine a heading by measuring ambient magnetic fields, such as the terrestrial magnetic field. The magnetometer **120**(**12**) may be worn or carried by the user **116**, mounted to the tote **118**, the user device **134**, and so forth. For example, the magnetometer **120**(**12**) in the user device **134** as worn by the user **116**(**1**) may act as a compass and provide information indicative of which way the user **116**(**1**) is facing.
(70) The sensors **120** may include other sensors **120**(S) as well. For example the other sensors **120**(S) may include proximity sensors, ultrasonic rangefinders, thermometers, barometric sensors, hygrometers, or biometric input devices including, but not limited to, fingerprint readers or palm scanners.
(71) The facility **102** may include one or more access points **210** configured to establish one or more wireless networks. The access points **210** may use Wi-Fi, near field communication (NFC), Bluetooth, or other technologies to establish wireless communications between a device and the network **202.** The wireless networks allow the devices to communicate with one or more of the inventory management system **122**, the sensors **120**, the user devices **134**, the RF tag **206**, a communication device of the tote **118**, or other devices.
(72) The output devices **130** may also be provided in the facility **102**. The output devices **130** are configured to generate signals which may be perceived by the user **116**.
(73) Haptic output devices **130**(**1**) are configured to provide a signal which results in a tactile sensation of the user **116.** The haptic output devices **130**(**1**) may use one or more mechanisms such as electrical stimulation or mechanical displacement to provide the signal. For example, the haptic output devices **130**(**1**) may be configured to generate a modulated electrical signal which produces an apparent tactile sensation in one or more fingers of the user **116**. In another example, the haptic output devices **130**(**1**) may comprise piezoelectric or rotary motor devices configured to provide a vibration which may be felt by the user **116**.
(74) One or more audio output devices **130**(**2**) are configured to provide acoustic output. The acoustic output includes one or more of infrasonic sound, audible sound, or ultrasonic sound. The audio output devices **130**(**2**) may use one or more mechanisms to generate the sound. These mechanisms may include, but are not limited to: voice coils, piezoelectric elements, magnetostrictive elements, or electrostatic elements, and so forth. For example, a piezoelectric buzzer or a speaker may be used to provide acoustic output.

(75) The display devices **130**(**3**) may be configured to provide output which may be seen by the user **116**, or detected by a light-sensitive detector such as an imaging sensor **120**(**1**) or light sensor **120**(**7**). The output may be monochrome or color. The display devices **130**(**3**) may be emissive, reflective, or both. An emissive display device **130**(**3**) is configured to emit light during operation. For example, a light emitting diode (LED) is an emissive visual display device **130**(**3**). In comparison, a reflective display device **130**(**3**) relies on ambient light to present an image. For example, an electrophoretic display is a reflective display device **130**(**3**). Backlights or front lights may be used to illuminate the reflective visual display device **130**(**3**) to provide visibility of the information in conditions where the ambient light levels are low.
(76) Mechanisms of the display devices **130**(**3**) may include liquid crystal displays **130**(**3**)(**1**), transparent organic light emitting diodes (LED) **130**(**3**)(**2**), electrophoretic displays **130**(**3**)(**3**), image projectors **130**(**3**)(**4**), or other display mechanisms **130**(**3**)(S). The other display mechanisms **130**(**3**)(S) may include, but are not limited to, micro-electromechanical systems (MEMS), spatial light modulators, electroluminescent displays, quantum dot displays, liquid crystal on silicon (LCOS) displays, cholesteric displays, interferometric displays, and so forth. These mechanisms are configured to emit light, modulate incident light emitted from another source, or both.
(77) The display devices **130**(**3**) may be configured to present images. For example, the display devices **130**(**3**) may comprise a pixel-addressable display. The image may comprise at least a two-dimensional array of pixels, or a vector representation of an at least two-dimensional image.
(78) In some implementations, the display devices **130**(**3**) may be configured to provide non-image data, such as text characters, colors, and so forth. For example, a segmented electrophoretic display, segmented LED, and so forth may be used to present information such as a SKU number. The display devices **130**(**3**) may also be configurable to vary the color of the text, such as using multicolor LED segments.
(79) In some implementations, display devices **130**(**3**) may be configurable to provide image or non-image output. For example, an electrophoretic display **130**(**3**)(**3**) with addressable pixels may be used to present images of text information, or all of the pixels may be set to a solid color to provide a colored panel.
(80) The output devices **130** may include hardware processors, memory, and other elements configured to present a user interface **132**. In one implementation, the display devices **130**(**3**) may be arranged along the edges of inventory locations **114**. The speech interaction system **124** may hear user speech **128** such as “where is that widget” and may be configured to present arrows on the display devices **130**(**3**) which point to the inventory location **114** which contains the “widget” item **104**.
(81) Other output devices **130**(T) may also be present. The other output devices **130**(T) may include lights, scent/odor dispensers, document printers, three-dimensional printers or fabrication equipment, and so forth. For example, the other output devices **130**(T) may include lights which are located on the inventory locations **114**, the totes **118**, and so forth. Continuing the example, the action of the speech interaction system **124** may comprise activating one or more lights on the tote **118** to navigate the user **116** through the facility **102** to a particular location.
(82) The inventory management system **122** may generate the user interface data which is then used by the output device **130**, the user devices **134**, or both to present a user interface **132**. The user interface **132** may be configured to stimulate one or more senses of the user **116**. For example, the user interface **132** may comprise visual, audible, and haptic output.
(83) FIG. **3** illustrates a block diagram **300** of the server **204**. The server **204** may be physically present at the facility **102**, may be accessible by the network **202**, or a combination of both. The server **204** does not require end-user knowledge of the physical location and configuration of the system that delivers the services. Common expressions associated with the server **204** may include “on-demand computing,” “software as a service (SaaS),” “platform computing,” “network-accessible platform,” “cloud services,” “data centers” and so forth. Services provided by the server **204** may be distributed across one or more physical or virtual devices.
(84) The server **204** may include one or more hardware processors **302** (processors) configured to execute one or more stored instructions. The processors **302** may comprise one or more cores. The server **204** may include one or more input/output (I/O) interface(s) **304** to allow the processor **302** or other portions of the server **204** to communicate with other devices. The I/O interfaces **304** may comprise inter-integrated circuit (I2C), serial peripheral interface bus (SPI), Universal Serial Bus (USB) as promulgated by the USB Implementers Forum, RS-232, and so forth.
(85) The I/O interface(s) **304** may couple to one or more I/O devices **306**. The I/O devices **306** may include input devices such as one or more of a keyboard, mouse, scanner, the sensors **120**, and so forth. The I/O devices **306** may also include output devices **130** such as one or more of a display, printer, audio speakers, and so forth. In some embodiments, the I/O devices **306** may be physically incorporated with the server **204** or may be externally placed.
(86) The server **204** may also include one or more communication interfaces **308**. The communication interfaces **308** are configured to provide communications between the server **204** and other devices, such as the sensors **120**, the user devices **134**, routers, the access points **210**, and so forth. The communication interfaces **308** may include devices configured to couple to personal area networks (PANs), wired and wireless local area networks (LANs), wired and wireless wide area networks (WANs), and so forth. For example, the communication interfaces **308** may include devices compatible with Ethernet, Wi-Fi™, and so forth.
(87) The server **204** may also include one or more busses or other internal communications hardware or software that allow for the transfer of data between the various modules and components of the server **204**.
(88) As shown in FIG. **3**, the server **204** includes one or more memories **310**. The memory **310** comprises one or more computer-readable storage media (“CRSM”). The CRSM may be any one or more of an electronic storage medium, a magnetic storage medium, an optical storage medium, a quantum storage medium, a mechanical computer storage medium, and so forth. The memory **310** provides storage of computer-readable instructions, data structures, program modules, and other data for the operation of the server **204**. A few example functional modules are shown stored in the memory **310**, although the same functionality may alternatively be implemented in hardware, firmware, or as a system on a chip (SOC).
(89) The memory **310** may include at least one operating system (OS) module **312**. The OS module **312** is configured to manage hardware resource devices such as the I/O interfaces **304**, the I/O devices **306**, the communication interfaces **308**, and provide various services to applications or modules executing on the processors **302**. The OS module **312** may implement a variant of the FreeBSD operating system as promulgated by the FreeBSD Project, other UNIX or UNIX-like variants, a variation of the Linux operating system as promulgated by Linus Torvalds, the Windows Server operating system from Microsoft Corporation of Redmond, Wash., and so forth.
(90) Also stored in the memory **310** may be one or more of the following modules. These modules may be executed as foreground applications, background tasks, daemons, and so forth.
(91) A communication module **314** may be configured to establish communications with one or more of the sensors **120**, one or more of the user devices **134**, other servers **204**, or other devices. The communications may be authenticated, encrypted, and so forth.
(92) The memory **310** may store an inventory management module **316**. The inventory management module **316** is configured to provide the inventory functions as described herein with regard to the inventory management system **122**. For example, the inventory management module **316** may track items **104** between different inventory locations **114**, to and from the totes **118**, and so forth.
(93) The inventory management module **316** may be configured to generate the user interface data which may be used by the output devices **130**, user devices **134**, and so forth to present the user interface **132**. The inventory management module **316** may include one or more of a data acquisition module **318**, a speech interaction module **320**, or a user interface module **322**.
(94) The data acquisition module **318** is configured to acquire input from one or more of the sensors **120**. For example, the data acquisition module **318** may be configured to receive image data **328**(**1**) from the imaging sensors **120**(**1**) and audio data **328**(**2**) generated by the microphones **120**(**5**). In some implementations, the data acquisition module **318** may analyze the sensor data **328** to generate additional information. For example, the data acquisition module **318** may use one or more techniques to localize or determine a position from which a sound emanates. These techniques may include beamforming using an array of a plurality of microphones **120**(**5**), time difference of arrival (TDOA), analysis of a phase of sounds received

by the array of microphones **120**(**5**), and so forth.
(95) The data acquisition module **318** may also be configured to acquire information from the user device **134**. In one implementation, the user device **134** may include an application module or operating system configured to provide information about messages sent or received by the user **116** of the user device **134**, application state such as whether a particular application is executing and if so what portion of the user interface **132** is being presented, and so forth. For example, the user device **134** may include an instant messaging application which is used to provide communication between users **116** of the facility **102** and other parties such as customers. The user device **134** in this example may be configured to provide to the data acquisition module **318** information indicative of messages sent or received by the user device **134**. The information acquired may be incorporated into the user context data **126** for use in processing the user speech **128** to perform one or more actions within the facility **102**.
(96) The speech interaction system **124** may be implemented as the speech interaction module **320**. The speech interaction module **320** processes the sensor data **328** to determine intent, meaning, actors, or other aspects of the user speech **128**. Once determined, particular actions may be selected and then initiated. These actions may include, but are not limited to, storing data, retrieving data, activating one or more mechanisms within the facility **102**, directing other users **116** of the facility **102**, and so forth.
(97) To facilitate the determination, the speech interaction module **320** may access the user context data **126**. The speech interaction module **320** processes user speech **128** which may not be specifically directed to the inventory management module **316**. Rather, the user speech **128** is incidental to the presence of the user **116** in the facility **102**. The user speech **128** may comprise the self-speak or muttering of the user **116** while alone, a conversation between the two users **116** in the facility **102**, a conversation between the user **116** in the facility **102** and another person elsewhere such as via telephone, and so forth. The natural language as expressed by the user **116** in the user speech **128** may be highly variable in terms of wordbase, lexicon, grammar, syntax, and so forth. For example, when the user **116** is specifically addressing a traditional speech recognition system the user **116** may speak with a particular syntax and grammar. In comparison, when the user **116** is speaking to another person a looser or more complicated syntax and grammar may be used, such as that involving slang, idiom, allusion, personal references, and so forth.
(98) To process the user speech **128**, the speech interaction module **320** may access the user context data **126** to determine a natural language understanding of the user speech **128**. The user context data **126** provides cues or information which may be used to determine wordbase, grammar, syntax, personal references, and so forth. For example, the user context data **126** may comprise information indicative of an item **104**(**1**) in the inventory location **114**(**1**) which the user **116**(**1**) is pointing at with an index finger. Based on this information, the speech interaction module **320** may process user speech **128** of “what's wrong with that?” to determine that the user **116** is inquiring about the status of the item **104**(**1**) in the inventory location **114**(**1**) and initiate an action which generates a user interface **132** which presents information about the status of the item **104**(**1**).
(99) The speech interaction module **320** may analyze the user speech **128** to determine whether the user speech **128** is directed to no one else (e.g., self-speech such as talking to oneself), to another user **116**, to an automated system, and so forth. The analysis may include assessment of attributes such as intonation, speaking rate, fluency, perplexity, and so forth to determine or suggest an intended recipient of the user speech **128**. For example, the intonation describes variation in spoken pitch as distinct from phonemes while the speaking rate indicates how fast the user **116** is generating communicative sounds. Perplexity may be indicative of relative complexity of the language models used to process the user speech **128**. For example, the perplexity may be indicative of, on average, how many different equally most probable words can follow any given word. Higher values of perplexity may be indicative of an increase in the information encoded in the user speech **128**.
(100) One or more thresholds may be set for these attributes and used to distinguish self-speech, conversation with another user **116**, interaction with another automated system, and so forth. For example, user speech **128** which exhibits average speaking rate of above 120 words per minute, variation in intonation exceeding 10%, and high perplexity may be designated as user-to-user speech and may be analyzed accordingly using user context data **126** of the users **116** involved. In comparison, the user speech **128** which exhibits an average speaking rate of between 80 and 120 words, has intonation with variability below 10%, and a high perplexity may be deemed to be self-talk, such as the user **116** speaking to himself.
(101) The user interface module **322** is configured to generate user interface data. The user interface data may be based at least in part on output from the speech interaction module **320**. For example, the speech interaction module **320** may retrieve information responsive to the user speech **128** and pass this information to the user interface module **322** for generation of user interface data and subsequent presentation of the user interface **132**. In some implementations, the user interface **132** may provide information associated with one or more of: contemporaneous user speech **128** or previous user speech **128**. For example, the user interface **132** may present visually on a display output device **130**(**3**) a listing of the last ten actions taken responsive to user speech **128**. The user interface **132** may also be configured to accept information from the user **116**, such as an indication as to whether the action taken was helpful. As described below, this input may be used to train the speech interaction module **320**.
(102) As described above, the user interface data is configured to provide the user interface **132** by way of one or more output devices **130**, user devices **134**, or a combination thereof. The user interface **132** may include one or more of haptic, audible, or visual stimuli. For example, the user interface **132** may be audible such that synthesized speech is generated by the user interface module **322** and presented by the audio output devices **130**(**2**).
(103) In some implementations the speech interaction module **320** may be unable to resolve the user speech **128** to a single action. For example, the user speech **128** as interpreted with regard to the user context data **126** may have two or more equally valid actions. In some implementations a single action may be selected, or one or more of the actions may be taken. For example, the speech interaction module **320** may be unable to resolve whether the user **116** is having trouble finding the item **104** “widget” on the shelf, or whether the user **116** would like to change a quantity of the widgets for picking. The speech interaction module **320** may be configured to take multiple actions, such as providing navigational cues to the user **116** to find the item **104** and presenting in the user interface **132** controls which allow for changing the quantity of the widgets on the pick list.
(104) Other modules **324** may also be present in the memory **310**. For example, an object recognition module may be configured to use data from one or more of the sensors **120** to identify an object such as the item **104**, the user **116**, the tote **118**, and so forth.
(105) The memory **310** may also include a data store **326** to store information. The data store **326** may use a flat file, database, linked list, tree, executable code, script, or other data structure to store the information. In some implementations, the data store **326** or a portion of the data store **326** may be distributed across one or more other devices including other servers **204**, network attached storage devices and so forth.
(106) The data store **326** may also include sensor data **328**. The sensor data **328** comprises information acquired from, or based on, the one or more sensors **120**. For example, the sensor data **328** may comprise three-dimensional information about an object in the facility **102**. As described above, the sensors **120** may include an imaging sensor **120**(**1**) which is configured to acquire one or more images. These images may be stored as image data **328**(**1**). The image data **328**(**1**) may comprise information descriptive of a plurality of picture elements or pixels. The microphones **120**(**5**) are configured to acquire audio data **328**(**2**) which is representative of sounds at the facility **102**. The audio data **328**(**2**) may include user speech **128**, movement sounds, equipment sounds, and so forth.
(107) In some implementations, the sensor data **328** may be acquired at least in part by the user devices **134**. For example, the data acquisition module **318** may receive image data **328**(**1**) and audio data **328**(**2**) from an imaging sensor **120**(**1**) and a microphone **120**(**5**) which are onboard a tablet computer of the user **116**.
(108) The data store **326** may also store the user context data **126**. In some implementations, a portion of the user context data **126** may be retrieved from other data stores or devices. The user context data **126** is configured to provide information about one or more of: operation of the facility **102**, data about the items **104**, the user **116** as an anonymous individual, the user **116** as an identified individual, data external to the facility **102** such as weather or delivery route information, and so forth.
(109) The user context data **126** may include physical layout data **330**. The physical layout data **330** provides a mapping of physical positions within the facility **102** of devices and objects such as the sensors **120**, inventory locations **114**, and so forth. For example, the physical layout data **330** may

indicate the coordinates within the facility **102** of an inventory location **114**, an RFID reader **120**(**8**) close to that inventory location **114**, and so forth. In some implementations, the inventory management module **316** may access the physical layout data **330** to determine a distance between two objects, such as the users **116**(**1**) and **116**(**2**).

(110) Item data **332** may also be included in the user context data **126**. The item data **332** comprises information associated with the items **104**. The information may include one or more inventory locations **114**, at which one or more of the items **104** are stored. The item data **332** may also include order data, SKU or other product identifier, price, quantity on hand, weight, expiration date, images of the item, detailed description information, ratings, ranking, and so forth. The inventory management module **316** may store information associated with inventory management functions in the item data **332**. The speech interaction module **320** may be configured to retrieve information about the items **104** which are mentioned in, or relevant to, the user speech **128**.

(111) The user context data **126** may also include historical activity data **334**, position data **336**, proximity data **338**, or gaze direction data **340**. The historical activity data **334** provides information about the user's **116** interactions with the facility **102** and the objects therein, such as the items **104**, the inventory locations **114**, other users **116**, the totes **118**, and so forth. For example, the historical activity data **334** may indicate a time of entry to the facility **102**, a route taken while in the facility **102**, a list of items **104** which have been picked or placed, and so forth. The historical activity data **334** may be associated with the user **116** which has been identified or unidentified. In some implementations, the historical activity data **334** may be retrieved, such as in the circumstance of an identified user **116**. For example, the user **116** may be identified upon entry to the facility **102**. After identification, historical activity data **334** associated with a user account of the user **116** may be retrieved from the data store **326**. The speech interaction module **320** may use the historical activity data **334** to ascertain what or whom the user **116** has interacted with in the facility **102**.

(112) The position data **336** comprises information indicative of the user's **116** position in the facility **102**. The position data **336** may be expressed in absolute terms or relative terms. Absolute terms may comprise a latitude, longitude, and altitude with respect to a geodetic reference point. Relative terms may include a position of 25.4 m along an x-axis and 75.2 m along a y-axis as designated by a floor plan of the facility **102**, 5.2 m from an inventory location **114** along the heading of 169°, and so forth. The position data **336** may be used by the speech interaction module **320** to determine user context data **126**. For example, based on the position data **336** in the facility, and using the information about inventory locations **114** and the corresponding items **104** therein, the speech interaction module **320** may determine the user context data **126** includes the items **104** which are nearby.

(113) The proximity data **338** provides information which is indicative of a spatial arrangement between the user **116** and another person. For example, the proximity data **338** may indicate that the user **116**(**1**) is within a threshold distance of the user **116**(**2**). In some implementations, the threshold distances may be specified for individual users **116**. The threshold distances may be static or dynamically adjusted. For example, the threshold distance may be configured to 3 m from users **116** which are unrelated, and 1 m from the users **116** which have a familial relationship. The speech interaction module **320** may use the proximity data **338** to process the user speech **128**. For example, the user **116**(**1**) may say the user speech **128** “what do you have to pick next?” Based at least in part on the proximity data **338** indicating that the user **116**(**2**) is within a threshold distance and no other users **116** are within the threshold distance, the speech interaction module **320** may perform several actions. These actions may include accessing information indicative of the user's **116**(**2**) pick list and presenting a user interface **132** which displays the information about the next item **104** on the user device **134**(**2**) associated with the user **116**(**2**).

(114) The gaze direction data **340** provides information indicative of a direction the user **116** is looking. In some implementations, the gaze direction data **340** may be determined by processing the image data **328**(**1**) of one or more eyes of the user **116**. Based at least in part on the position data **336** of the user **116** and the gaze direction data **340**, it may be determined what object the user **116** is looking at. By using this information, the speech interaction module **320** may be able to disambiguate user speech **128**. For example, the user **116**(**1**) may be a supervisor who utters the user speech **128** “please get me another tote” while looking at the user **116**(**3**) who is several meters away. The speech interaction module **320** may use the gaze direction data **340** to determine that the user **116**(**1**) is addressing the user **116**(**3**), and may perform the actions of providing a user interface **132** which is indicative of a closest tote **118**.

(115) In other implementations, other directional data associated with the user **116** may be determined. For example, the directional data may indicate overall orientation of the user's **116** body, the direction of their head, and so forth.

(116) The user context data **126** may also include user data **342**. The user data **342** comprises information which is specific to an identified user **116**. Once the user **116** has been identified, the user data **342** associated with that identity may be retrieved. The user data **342** may include demographic data **344**, relationship data **346**, user preferences **348**, navigation path data **350**, access permissions **352**, or other information.

(117) The demographic data **344** may comprise information indicative of the user's **116** skills, address, contact information, age, and so forth. The relationship data **346** provides information indicative of an association between the user **116** and other users or people. For example, the relationship data **346** may indicate the users **116**(**1**) and **116**(**2**) are supervisor and subordinate, respectively. In another implementation, the relationship data **346** may indicate that the users **116**(**1**) and **116**(**2**) share a familial or marital relationship, such as being spouses or siblings. The relationship data **346** may be used by the inventory management module **316** in various ways. For example, the relationship data **346** may be accessed to prevent having related individuals crosscheck one another's picking of a high-value item **104**. The speech interaction module **320** may use the relationship data **346** to disambiguate, determine meaning, or otherwise process the user speech **128**.

(118) The user preferences **348** may include information such as preferred colors for visual display, preferences as to the timbre of a synthesized voice, preference for audible output over visual output, threshold distances for different actions, and so forth. For example, the user preference **348** may include threshold distance data indicating that the user **116**(**1**) prefers a visual user interface **132** when other users **116** are within a 3 m threshold distance and an audible user interface **132** when the other users are beyond the 3 m threshold distance. The thresholds may be modified based on the physical layout data **330**. For example, the threshold distance may be within a “line of sight”, such as between users **116** which are within the same aisle **112** as compared to users **116** which are in adjacent aisles.

(119) The navigation path data **350** provides information indicative of an actual, scheduled, or predicted path of the user **116** through the facility **102**. For example, the historical navigation path data **350** may comprise a time series of waypoints of the user **116** within the facility **102** over a period of elapsed time. The waypoints may specify a location within the facility **102**. The historical navigation path data **350** provides information showing where the user **116** has been within the facility **102**.

(120) Scheduled navigation path data **350** provides information as to what path the user **116** is being directed to follow through the facility **102**. For example, the scheduled navigation path data **350** may be generated using a list of items **104** to be picked by the user **116**. Based at least in part on the physical arrangement of the inventory locations **114** associated with those items **104** in the order as retrieved from the physical layout data **330**, a picking path may be specified. In some implementations, the scheduled navigation path data **350** may also comprise a time series indicating estimated positions within the facility **102** at particular times. The routing of the scheduled navigation path data **350** may be configured to optimize one or more parameters of the user's **116** travel through the facility **102**. For example, the scheduled navigation path data **350** may be configured to minimize walking distance, reduce congestion at a particular inventory location **114** relative to other users **116**, minimize use of other material handling equipment such as pallet jacks or forklifts, and so forth.

(121) Predicted navigation path data **350** provides an estimation as to what path the user **116** will follow through the facility **102** prospectively. In some implementations, the predicted navigation path data **350** may also comprise a time series indicating estimated positions within the facility **102** at particular times. For example, the predicted navigation path data **350** may be based on the historical navigation path data **350** of one or more users **116**. Continuing the example, the historical navigation path data **350** may indicate that a significant percentage of the users **116** that have gone past a particular aisle **112**, enter that aisle **112** even when no items **104** within the aisle **112** are scheduled for picking. In this example, predicted navigation

path data **350** for a particular user **116** may include the assumption that this particular user **116** will also enter that aisle **112.**
(122) In some implementations, certain portions of the facility **102** may be blanked out such that position data **336** is not collected, or may have navigation path data **350** associated with those portions redacted from the navigation path data **350**. For example, movements within a break room or a restroom are not tracked.
(123) The speech interaction module **320** may use the navigation path data **350** to establish where the user **116** may have been, or may be going next. For example, the user speech **128** may be the user **116**(**1**) saying “my tote is so heavy” aloud. The speech interaction module **320** may determine the inventory location **114** which the user **116**(**1**) is predicted to visit, and may dispatch another user **116**(**2**) to meet the first user **116**(**1**) and help with manipulating the heavily laden tote **118**.
(124) The user data **342** may also include access permissions **352** which specify what actions may be performed by a particular user **116** or group of users **116**. For example, the user **116**(**1**) may be assigned access permissions **352** commensurate with their responsibilities as a supervisor. These access permissions **352** may allow the supervisor user **116**(**1**) to see detailed information about the tasks currently being performed by other users **116**, allow for repositioning of inventory locations **114**, and so forth.
(125) In another example, the user **116**(**1**) may have set access permissions **352** restricting dissemination of user data **342** to other users **116**. The access permissions **352** may be specified in terms of which users **116** may retrieve user data **342**, what user data **342** the particular user **116** chooses to allow others to access, or a combination thereof.
(126) Individual users **116** or groups of users **116** may selectively provide user data **342** for use by the inventory management system **122**, or may authorize collection of the user data **342** during use of the facility **102** or access to user data **342** obtained from other systems. For example, a user **116** may opt-in to collection of the user data **342** to receive enhanced services while using the facility **102**.
(127) The data store **326** may also store user interface data **354**. The user interface data **354** may comprise commands, instructions, tags, markup language, images, color values, text, or other data. For example, the user interface data **354** may be expressed as hypertext markup language (HTML), cascading style sheets (CSS), JavaScript, and so forth. One or more output devices **130**, user devices **134**, or a combination thereof are configured to use the user interface data **354** to present the user interface **132** which may be perceived by the user **116**. The user interface **132** may include one or more elements including visual, haptic, audible, olfactory, and so forth. For example, the user interface **132** may be a graphical user interface, audible user interface, haptic user interface, or a combination thereof.
(128) The data store **326** may also include other data **356**. For example, the other data **356** may include information such as configuration files, weather information, loading dock status, tote **118** availability, and so forth.
(129) The server **204** may also include a power supply **358**. The power supply **358** is configured to provide electrical power suitable for operating the components in the server **204**.
(130) FIG. **4** illustrates a block diagram **400** of the tote **118**, according to some implementations. The tote **118** may include an RF tag **206**. The RF tag **206** may be affixed to, integral with, or is otherwise associated with the tote **118**. In some implementations, the tote **118** may have identifiers, tags, or other indicia thereupon. For example, a machine-readable optical code, such as a barcode, may be affixed to a side of the tote **118**.
(131) The tote **118** may include one or more hardware processors **402** (processors) configured to execute one or more stored instructions. The processors **402** may comprise one or more cores. The tote **118** may include one or more I/O interface(s) **404** to allow the processor **402** or other portions of the tote **118** to communicate with other devices. The I/O interfaces **404** may include I2C, SPI, USB, RS-232, and so forth.
(132) The I/O interface(s) **404** may couple to one or more I/O devices **406**. The I/O devices **406** may include one or more of the input devices such as the sensors **120**. As described above, the sensors **120** may include imaging sensors **120**(**1**), weight sensors **120**(**6**), RFID readers **120**(**8**), and so forth. The I/O devices **406** may also include output devices **130** such as haptic output devices **130**(**1**), audio output devices **130**(**2**), display devices **130**(**3**), and so forth. For example, the tote **118** may include other display devices **130**(T) such as lights which may be activated to provide information to the user **116**. In some implementations, input and output devices may be combined. For example, a touchscreen display may incorporate a touch sensor **120**(**4**) and a display device **130**(**3**). In some embodiments, the I/O devices **406** may be physically incorporated with the tote **118** or may be externally placed.
(133) The tote **118** may also include one or more communication interfaces **408**. The communication interfaces **408** are configured to provide communications between the tote **118** and other devices, such as other totes **118**, user devices **134**, routers, access points **210**, the servers **204**, and so forth. The communication interfaces **408** may include devices configured to couple to PANs, LANs, WANs, and so forth. For example, the communication interfaces **408** may include devices compatible with Ethernet, Wi-Fi, Bluetooth, ZigBee, and so forth.
(134) The tote **118** may also include one or more busses or other internal communications hardware or software that allow for the transfer of data between the various modules and components of the tote **118**.
(135) As shown in FIG. **4**, the tote **118** includes one or more memories **410**. The memory **410** comprises one or more CRSM as described above. The memory **410** provides storage of computer readable instructions, data structures, program modules and other data for the operation of the tote **118**. A few example functional modules are shown stored in the memory **410**, although the same functionality may alternatively be implemented in hardware, firmware, or as a SOC.
(136) The memory **410** may include at least one OS module **412**. The OS module **412** is configured to manage hardware resource devices such as the I/O interfaces **404**, the I/O devices **406**, the communication interfaces **408**, and provide various services to applications or modules executing on the processors **402**. The OS module **412** may implement a variant of the FreeBSD operating system as promulgated by the FreeBSD Project, other UNIX or UNIX-like variants, a variation of the Linux operating system, such as Android as promulgated by Google, Inc. of Mountain View, Calif. Other OS modules **412** may be used, such as the Windows operating system from Microsoft Corporation of Redmond, Wash., the LynxOS from LynuxWorks of San Jose, Calif., and so forth.
(137) Also stored in the memory **410** may be one or more of the following modules. These modules may be executed as foreground applications, background tasks, daemons, and so forth.
(138) A communication module **414** may be configured to establish communications with one or more of the sensors **120**, other totes **118**, the user devices **134**, the servers **204**, or other devices. The communications may be authenticated, encrypted, and so forth.
(139) The memory **410** may also store a tote item tracking module **416**. The tote item tracking module **416** is configured to maintain a list of items **104**, which are associated with the tote **118**. For example, the tote item tracking module **416** may receive input from a user **116** by way of a touch screen display with which the user **116** may enter information indicative of the item **104** placed in the tote **118**. In another example, the tote item tracking module **416** may receive input from the I/O devices **406**, such as the weight sensor **120**(**6**) and an RFID or NFC reader **120**(**8**). The tote item tracking module **416** may send the list of items **104** to the inventory management system **122**. The tote item tracking module **416** may also be configured to receive information from the inventory management system **122**. For example, a list of items **104** to be picked may be presented within a user interface **132** on the display device **130**(**3**) of the tote **118**.
(140) The memory **410** may include a display module **418**. The display module **418** may be configured to present information, such as received from the one or more servers **204** or generated onboard the tote **118**, using one or more of the output devices **130**. For example, the display module **418** may be configured to receive user interface data **354** provided by the user interface module **322**. By processing user interface data **354**, the user interface **132** may be presented to the user **116** by way of the output devices **130** of the tote **118.** For example, the user interface **132** may include haptic output from the haptic output devices **130**(**1**), audio output from the audio output devices **130**(**2**), images presented on the display output devices **130**(**3**), activation of lights or other output devices **130**(T), or a combination thereof. Continuing the example, the user interface data **354** may be generated by the speech interaction module **320** in response to processing of the user speech **128**.

(141) Other modules **420** may also be stored within the memory **410**. In one implementation, a data handler module may be configured to generate sensor data **328**. For example, an imaging sensor **120**(**1**) onboard the tote **118** may acquire image data **328**(**1**) and one or more microphones **120**(**5**) onboard the tote **118** may acquire audio data **328**(**2**). The sensor data **328**, or information based thereon, may be provided to the data acquisition module **318**.
(142) The other modules **420** may also include a user authentication module which may be configured to receive input and authenticate or identify a particular user **116**. For example, the user **116** may enter a personal identification number or may provide a fingerprint to the fingerprint reader to establish their identity.
(143) The memory **410** may also include a data store **422** to store information. The data store **422** may use a flat file, database, linked list, tree, executable code, script, or other data structure to store the information. In some implementations, the data store **422** or a portion of the data store **422** may be distributed across one or more other devices including the servers **204**, other totes **118**, network attached storage devices and so forth.
(144) The data store **422** may store a tote item identifier list **424**. The tote item identifier list **424** may comprise data indicating one or more items **104** associated with the tote **118**. For example, the tote item identifier list **424** may indicate the items **104** which are present in the tote **118**. The tote item tracking module **416** may generate or otherwise maintain a tote item identifier list **424**.
(145) A unique identifier **426** may also be stored in the memory **410**. In some implementations, the unique identifier **426** may be stored in rewritable memory, write-once-read-only memory, and so forth. For example, the unique identifier **426** may be burned into a one-time programmable non-volatile memory, such as a programmable read-only memory (PROM). In some implementations, the unique identifier **426** may be part of a communication interface **408**. For example, the unique identifier **426** may comprise a media access control address associated with a Bluetooth interface. In some implementations, the user interface module **322** may use the unique identifier **426** to determine which tote **118** to generate the user interface **132** upon, or to determine a source for the sensor data **328**.
(146) The data store **422** may also store sensor data **328**. The sensor data **328** may be acquired from the sensors **120** onboard the tote **118**. The user interface data **354** received by the tote **118** may also be stored in the data store **422**.
(147) Other data **428** may also be stored within the data store **422**. For example, tote configuration settings, user interface preferences, and so forth may also be stored.
(148) The tote **118** may also include a power supply **430**. The power supply **430** is configured to provide electrical power suitable for operating the components in the tote **118**. The power supply **430** may comprise one or more of photovoltaic cells, batteries, wireless power receivers, fuel cells, capacitors, and so forth.
(149) FIG. **5** illustrates a block diagram **500** of the user device **134**, according to some implementations. The user device **134** may comprise a tablet computer, smartphone, notebook computer, data access terminal, wearable computing device, and so forth. The user device **134** may include one or more hardware processors **502** (processors) configured to execute one or more stored instructions. The processors **502** may comprise one or more cores. The user device **134** may include one or more I/O interface(s) **504** to allow the processor **502** or other portions of the user device **134** to communicate with other devices. The I/O interfaces **504** may include I2C, SPI, USB, RS-232, and so forth.
(150) The I/O interface(s) **504** may couple to one or more I/O devices **506**. The I/O devices **506** may include one or more of the input devices such as the sensors **120**. As described above, the sensors **120** may include imaging sensors **120**(**1**), 3D sensors **120**(**2**), buttons **120**(**3**), touch sensors **120**(**4**), microphones **120**(**5**), weight sensors **120**(**6**), light sensors **120**(**7**), RFID readers **120**(**8**), accelerometers **120**(**10**), gyroscopes **120**(**11**), magnetometers **120**(**12**), and so forth.
(151) The I/O devices **506** may also include one or more of haptic output devices **130**(**1**), audio output devices **130**(**2**), display devices **130**(**3**), and so forth. For example, the display devices **130**(**3**) may be configured in a head-mounted display, such that the field of vision of the user **116** includes the display device **130**(**3**) as well as one or more of the real-world objects in the facility **102**. In some implementations, input and output devices may be combined. In some embodiments, the I/O devices **506** may be physically incorporated with the user device **134** or may be externally placed.
(152) The user device **134** may also include one or more communication interfaces **508**. The communication interfaces **508** are configured to provide communications between the user device **134** and other devices, such as other user devices **134**, totes **118**, routers, access points **210**, the servers **204**, and so forth. The communication interfaces **508** may include devices configured to couple to PANs, LANs, WANs, and so forth. For example, the communication interfaces **508** may include devices compatible with Wi-Fi, Bluetooth, ZigBee, and so forth.
(153) The user device **134** may also include one or more busses or other internal communications hardware or software that allow for the transfer of data between the various modules and components of the user device **134**.
(154) As shown in FIG. **5**, the user device **134** includes one or more memories **510**. The memory **510** comprises one or more CRSM as described above. The memory **510** provides storage of computer readable instructions, data structures, program modules and other data for the operation of the user device **134**. A few example functional modules are shown stored in the memory **510**, although the same functionality may alternatively be implemented in hardware, firmware, or as a SOC.
(155) The memory **510** may include at least one OS module **512**. The OS module **512** is configured to manage hardware resource devices such as the I/O interfaces **504**, the I/O devices **506**, the communication interfaces **508**, and provide various services to applications or modules executing on the processors **502**. The OS module **512** may implement a variant of the FreeBSD operating system as promulgated by the FreeBSD Project, other UNIX or UNIX-like variants, a variation of the Linux operating system, such as Android as promulgated by Google, Inc. of Mountain View, Calif. Other OS modules **512** may be used, such as the Windows operating system from Microsoft Corporation of Redmond, Wash., the LynxOS from LynuxWorks of San Jose, Calif., and so forth.
(156) Also stored in the memory **510** may be one or more of the following modules. These modules may be executed as foreground applications, background tasks, daemons, and so forth.
(157) A communication module **514** may be configured to establish communications with one or more of the sensors **120**, user devices **134**, the servers **204**, or other devices. The communications may be authenticated, encrypted, and so forth.
(158) The memory **510** may include a display module **516**. The display module **516** may be configured to parse the user interface data **354** and present the user interface **132** using one or more of the output devices **130** which are onboard or communicatively coupled to the user device **134**. For example, the display module **516** may include an HTML rendering engine configured to process the HTML in the user interface data **354**. The display module **516** may also be configured to modify the user interface **132** based at least in part on sensor data **328** acquired by sensors **120** onboard or communicatively coupled to the user device **134**. For example, the display module **516** may be configured to provide some object recognition and tracking, such that user interface elements presented by the display device **130**(**3**) follow the motion of the tracked object, such as the user **116** which is in the field of view.
(159) Other modules **518** may also be stored within the memory **510**. In one implementation, a data handler module may be configured to generate sensor data **328**. For example, the data handler module may be configured to acquire audio data **328**(**2**) from one or more microphones **120**(**5**) and provide that audio data **328**(**2**) to the data acquisition module **318** of the server **204**.
(160) The other modules **518** may also include a facial detection module which may be configured to determine a presence of a human face within the image data **328**(**1**) acquired by the onboard imaging sensor **120**(**1**). The image of the face may then be sent to the server **204** for facial recognition to determine who is present in the image.
(161) The memory **510** may also include a data store **520** to store information. The data store **520** may use a flat file, database, linked list, tree, executable code, script, or other data structure to store the information. In some implementations, the data store **520** or a portion of the data store **520** may be distributed across one or more other devices including the servers **204**, other user devices **134**, network attached storage devices and so forth.

(162) The data store **520** may store sensor data **328** as acquired from the sensors **120** onboard the user device **134** or communicatively coupled thereto. For example, the sensor data **328** may include image data **328**(**1**) acquired for an onboard imaging sensor **120**(**1**) of the user device **134**. The user interface data **354** received by the user device **134** may also be stored in the data store **520**. Other data **522** may also be stored within the data store **520**. For example, device configuration settings, user interface preferences, and so forth may also be stored.
(163) The user device **134** may also include a power supply **524**. The power supply **524** is configured to provide electrical power suitable for operating the components in the user device **134**. The power supply **524** may comprise one or more of photovoltaic cells, batteries, wireless power receivers, fuel cells, capacitors, and so forth.
(164) FIG. **6** illustrates a side view **600** of an overhead imaging sensor **120**(**1**) acquiring an image of the users **116**, the tote **118**, and other objects. In some implementations, the facility **102** may include one or more sensors **120** which are configured to acquire an image from an overhead vantage point or at other positions within the facility **102**. The sensors **120** may include, but are not limited to, one or more of the imaging sensors **120**(**1**), the 3D sensors **120**(**2**), the microphones **120**(**5**), the RFID readers **120**(**8**), the RF receivers **120**(**9**), and so forth. The sensor data **328** acquired by the sensors **120** may be used to generate the user context data **126**.
(165) In this illustration, one of the sensors **120** comprises an imaging sensor **120**(**1**) which is configured to generate image data **328**(**1**). A field of view **602** of the imaging sensor **120**(**1**) depicted here includes the users **116**(**1**) and **116**(**2**) and the tote **118**. The image data **328**(**1**) may be provided to the inventory management system **122**. For example, the inventory management module **316** executing on the server **204** may process the image data **328**(**1**) to determine at least a portion of the user context data **126**, such as the position data **336** for the users **116**(**1**) and **116**(**2**).
(166) The one or more sensors **120** may also include an array of microphones **120**(**5**) configured to generate audio data **328**(**2**). The array of microphones **120**(**5**) may be arranged in a one, two, or three-dimensional arrangement. For example, a three-dimensional arrangement may comprise microphones **120**(**5**) arranged lengthwise along an aisle **112** at ceiling height and also lengthwise down the aisle on middle shelves of the inventory locations **114** for that aisle **112**.
(167) The array of microphones **120**(**5**) may be used to localize or determine a position within the space of a particular sound. As described above, the localization may use time difference of arrival (TDOA), phase differences, amplitude comparison, beamforming, or other techniques to determine the position. The position may be determined along a single dimension, along two dimensions, or in three-dimensional space. For example, the position may be determined as a position lengthwise along the aisle **112** as a single dimension. Determination of the position data **336** may be based at least in part on the localization information provided by the array of microphones **120**(**5**). For example, the localized position of the user speech **128** which is associated with the user **116**(**1**) may be combined with position information generated from the image data **328**(**1**) to provide the position data **336**.
(168) In some implementations, the data acquisition module **316** may be configured to generate user context data **126** using information from the plurality of sensors **120**. For example, image data **328**(**1**) may indicate that the user **116**(**1**) is standing alone and speaking based on facial motions of the user **116**(**1**) and an absence of other human faces. The audio data **328**(**2**) may comprise the user speech **128**. RF data from the RF receiver **120**(**9**) as well as information provided by the user device **134** may be indicative of an active cell phone call. Furthermore, the information provided by the user device **134** may indicate that the user **116**(**1**) is talking to the user **116**(**2**) who is designated by the relationship data **346** as being a subordinate of the user **116**(**1**). Based on this user context data **126**, the speech interaction module **320** may be able to derive meaning and perform one or more actions responsive to the user speech **128** which comprises the half of the conversation uttered by the user **116**(**1**). Thus, the speech interaction module **320** may be configured to initiate actions responsive to the user speech **128** so long as at least one of the participants is present or otherwise in communication with the facility **102**.
(169) As described here, the tote **118** may also include one or more output devices **130**, one or more sensors **120**, and so forth. Similarly the user device **134** may include one or more output devices **130**, one or more sensors **120**, and so forth. The data acquisition module **318** may be configured to access the data provided by these sensors **120** while the user interface module **322** may be configured to generate user interface data **354** configured to provide the user interface **132** by way of the output devices **130**.
(170) FIG. **7** illustrates a user interface **700** for setting how the facility **102** interacts with the user **116**, according to some implementations. One or more of the output devices **130** may present the user interface **700**.
(171) The user interface **700** may allow the user **116** to configure or otherwise set one or more of the user preferences **348**. The user information section **702** provides information about the user **116** who is presently accessing the user interface **700** and their account information.
(172) A user localization section **704** allows the user to view, set, or modify information such as a current location, preferred wordbase, and so forth. For example, as shown here the user **116** has a current location of North America indicative of the facility **102** being located within the United States. The preferred word base, in comparison, is that of the United Kingdom. Continuing the example, the user **116** may be a native of the United Kingdom. The wordbase may comprise words and their pronunciations, associated meanings, grammar, and so forth which may have particular regional, cultural, or other significance. For example, a United Kingdom wordbase may indicate that the word “torch” is equivalent to the American word for “flashlight”. The wordbase may be determined manually such as by input from the user **116**, or may be automatically determined based on analysis of the user speech **128** by the user **116**. In some implementations the word base may comprise a data structure which specifies interconnections or relationships between one or more lexicons, grammars, semantic data, and so forth.
(173) The user specific keywords section **706** allows the user to view, set, or modify particular key words or phrases. Keywords or phrases may be associated with particular actions, intents, or states of the user **116**. The association of particular keywords to the user **116**, groups of users **116**, categories of users **116**, and so forth, may be automatically determined by the speech interaction module **320**. For example, a grunt noise may be associated with frustration of a particular user **116**. In other implementations, the user interface **700** may allow the user **116** to specify that the grunt conveys this particular meaning. Responsive to this meaning, the speech interaction module **320** may be configured to provide additional information associated with a task assigned to the user **116** upon processing of the user speech **128** which includes a grunt.
(174) A user specific settings section **708** allows the user to specify other aspects of operation of the speech interaction module **320** or the inventory management module **316**. For example, the user **116** may give permission for the speech interaction module **320** to continuously monitor the audio data **328**(**2**) at all times. The user **116** may provide specific limitations such as including or excluding telephone calls, including or excluding conversations with children, and so forth. Other settings may include enabling the user gaze tracking, enabling operation with an application on the user device **134**, and so forth.
(175) The user specific settings section **708** may also allow the user **116** to specify whether information provided in the user interface **132** is to be personalized. Personalized information may include the user data **342**. For example, a non-personalized user interface **132** may present information such as a SKU and cost of an item **104**. In comparison, a personalized user interface **132** may present information such as a last quantity purchased by that user **116**.
(176) The user specific settings section **708** may allow the user **116** to specify whether audible output is to be provided, and if so whether to limit that audible output based on proximity data **338** or other parameters. Similarly, the user **116** may specify whether visual output is to be provided, and if so what limits with regard to proximity data **338** are to be applied.
(177) The user **116** may also specify user specific settings section **708** which enables the speech interaction module **322** to engage in an interactive conversation. For example, the user speech **128** may result in retrieval of information about an item **104**. The speech interaction module **320** may operate in conjunction with the user interface module **322** to generate the user interface **132** which presents synthesized speech or a visual prompt asking the user **116** for permission to present the retrieved information when this option is enabled. In another example, the speech interaction module **320** may operate in conjunction with the user interface module **322** to ask the user **116** questions or to provide additional information to

disambiguate or confirm an action. When not enabled, the system may generate and present the user interface **132** automatically.

(178) The user specific settings section **708** may also enable the speech interaction module **320** to learn from the interactions with the user **116**. For example, one or more machine learning techniques may be employed to process speech and determine which actions to provide for particular users **116**, all users **116**, or a combination thereof. For example, the speech interaction module **320** may process user speech **128** using a neural network to recognize text and determine an action associated therewith.

(179) Feedback may be provided to the machine learning system by monitoring further user speech **128**, user context data **126**, or a combination thereof. For example, subsequent user speech **128** such as “thank you”, “great,” or “that's helpful” may provide data which reinforces that the action determined by the speech interaction module **320** was appropriate with respect to the user speech **128**. In comparison, a user speech **128** such as “that's wrong” or “nope” may provide data which deprecates the action.

(180) Similarly, subsequent actions by the user **116** following an action by the inventory management system **122** may provide feedback to the speech interaction module **320**. For example, the action of moving the item **104**(**1**) close to the user **116**, who then subsequently picks up the item **104**(**1**), may reinforce that the user **116** wanted the item **104**(**1**) and that the action of moving the item **104**(**1**) was useful. In another example, the action of the user **116** moving toward the item **104**, or the inventory location **114** associated with the item **104**, provides feedback which may be analyzed. Similarly, user input from the user interface **132** may also be used to provide feedback to the speech interaction module **320**.

(181) Other sections which are not depicted here may be provided in the user interface **700**. For example, an access permission section may allow the user **116** to specify one or more of the access permissions **352**.

(182) FIG. **8** illustrates a scenario **800** of a single user **116** uttering user speech **128** and the system **100** responding to this, according to some implementations. As described above, the facility **102** may include one or more sensors **120** such as microphones **120**(**5**) and one or more output devices **130** such as audio output devices **130**(**2**). In this illustration, a single user **116** is shown in a portion of the facility **102** which includes an inventory location **114** holding various items **104**. While working by himself, the user **116** utters user speech **128** such as “widgets, widgets, widgets, (hmmm)”. The speech interaction module **320**, based on the user context data **126**, may interpret the user speech **128** as calling for the action of providing information to the user **116** about the location of the items **104** known as widgets.

(183) As illustrated here, the audio output device **130**(**2**) provides a user interface **132**(**1**) comprising synthesized audio saying “Allow me to help you. The widgets on your pick list are located to your left on rack A shelf 2.” Another user interface **132**(**2**) comprising text presented visually reading “Widgets are on rack A shelf 2” and presented on a display output device **130**(**3**) may be presented instead of, or in addition to, the audible user interface **132**(**1**). In some implementations the user **116** may be advised that the speech interaction system **124** has determined a suggested action or is ready to perform an action by a chime, haptic output, change in illumination of a light, and so forth. For example, the speakers on the tote **118** may chime while lights on the tote **118** may blink. The action may be performed, or may be held pending approval or assent by the user **116** to perform the action.

(184) As illustrated in this example, the user **116** has not explicitly engaged the system for operation. Indeed, the utterance of the user speech **128** may be accidental or unintentional and may lack a defined grammar or syntax. However, based on the user context data **126** which includes information about the pick list of items associated with the user **116**, as well as the item data **332**, the physical layout data **330**, the user position data **336**, and so forth, the speech interaction module **320** is able to provide useful information to the user **116** to facilitate performance of the user's **116** tasks.

(185) FIG. **9** illustrates a scenario **900** in which the system **100** responds to user speech **128** between two users **116**(**1**) and **116**(**2**) of the facility **102**, according to some implementations. As described above, the facility **102** may include one or more sensors **120** such as microphones **120**(**5**) and one or more output devices **130** such as audio output devices **130**(**2**). In this illustration, two users **116**(**1**) and **116**(**2**) are shown in a portion of the facility **102** which includes an inventory location **114** holding various items **104**. While working together, the first user **116**(**1**) says aloud to the second user **116**(**2**) “Bob, it seems like these widgets have gotten more expensive.” In terms of wording and inflection, this user speech **128** is declarative. However, the speech interaction module **320**, based on the user context data **126**, retrieves information about cost changes associated with the item **104** which is named “widget” and determines that an appropriate action is to present this information. The speech interaction module **320** may then provide this information to the user interface module **322** which generates user interface data **354** suitable for providing one or more of an audible user interface **132**(**1**) or a visual user interface **132**(**2**). The user **116**(**1**) may provide an affirmation that the user interface **132** providing information was of use by issuing further user speech **128** such as “thank you”.

(186) In another implementation not illustrated here, the user **116**(**1**) may be carrying on the conversation with a person or system not present at that position within the facility **102** or not in the facility **102** at all. For example, the user **116**(**1**) may be conversing with the user **116**(**2**) via a two-way radio, cell phone, or other audio communication device. In another example, the user **116**(**1**) may be providing verbal input to another system, such as providing instructions or a query to an intelligent personal assistant application executing at least in part on the user device **134**.

(187) Illustrative Processes

(188) FIG. **10** depicts a flow diagram **1000** of a process for performing actions based on speech between users **116** of the facility **102**, according to some implementations. In some implementations, the process may be performed at least in part by the inventory management module **316**.

(189) Block **1002** acquires images within the facility **102**. For example, the imaging sensors **120**(**1**) may generate image data **328**(**1**).

(190) Block **1004** acquires audio within the facility **102**. For example, the microphones **120**(**5**) may generate the audio data **328**(**2**).

(191) Block **1006** identifies a first user **116**(**1**) with the acquired images. For example, the image data **328**(**1**) may be processed using one or more techniques to determine identity of the first user **116**(**1**). These techniques may include one or more of facial recognition, clothing recognition, gait recognition, and so forth. In other implementations, other techniques may be used instead of or in addition to identification by way of the image data **328**(**1**). For example, a user **116** may be identified by way of an RFID tag, manual entry of credentials at an input device, and so forth.

(192) Facial recognition may include analyzing facial characteristics which are indicative of one or more facial features in an image, three-dimensional data, or both. The facial features include measurements of, or comparisons between, facial fiducials or ordinal points. The facial features may include eyes, mouth, lips, nose, chin, ears, face width, skin texture, three-dimensional shape of the face, presence of eyeglasses, and so forth. In some implementations the facial characteristics may include facial metrics. The facial metrics indicate various ratios of relative sizes and spacing of the facial features. For example, the facial metrics may include ratio of interpupillary distance to facial width, ratio of eye width to nose width, and so forth. In some implementations the facial characteristics may comprise a set of eigenvectors by using principal component analysis (PCA) on a set of images. These eigenvectors as descriptive of a human face may be known as “eigenfaces” or “eigenimages”. In one implementation the facial recognition described in this disclosure may be performed at least in part using one or more tools available in the OpenCV library as developed by Intel Corporation of Santa Clara, Calif., Willow Garage of Menlo Park, Calif., and Itseez of Nizhny Novgorod, Russia, with information available at www.opencv.org. In other implementations, other techniques may be used to recognize faces. Previously stored registration data may associate particular facial characteristics with a particular identity, such as represented by a user account. For example, the particular pattern of eigenvectors in the image may be sought in the previously stored data, and matches within a threshold tolerance may be determined to indicate identity. The facial characteristics may be used to identify the user **116**, or to distinguish one user **116** from another.

(193) Clothing recognition analyzes images to determine what articles of clothing, ornamentation, and so forth the user **116** is wearing or carrying. Skin and hair detection algorithms may be used to classify portions of the image which are associated with the user's **116** skin or hair. Items which are not skin and hair may be classified into various types of articles of clothing such as shirts, hats, pants, bags, and so forth. The articles of clothing may be classified according to function, position, manufacturer, and so forth. Classification may be based on clothing color, texture, shape, position on the user **116**, and so forth. For example, classification may designate an article of clothing worn on the torso as a “blouse” while color or pattern

information may be used to determine a particular designer or manufacturer. The determination of the article of clothing may use a comparison of information from the images with previously stored data. Continuing the example, the pattern of the blouse may have been previously stored along with information indicative of the designer or manufacturer.
(194) In some implementations, identification of the user **116** may be based on the particular combination of classified articles of clothing. For example, the user **116**(**2**) may be distinguished from the user **116**(**3**) based at least in part on the user **116**(**2**) wearing a hat and a red shirt while the user **116**(**3**) is not wearing a hat and is wearing a blue shirt. The clothing may be used to identify the user **116**, or distinguish one user **116** from another.
(195) Gait recognition analyzes images, three-dimensional data, or both to assess how a user **116** moves over time. Gait comprises a recognizable pattern of movement of the user's **116** body which is affected by height, age, and other factors. Gait recognition may analyze the relative position and motion of limbs of the user **116**. Limbs may include one or more arms, legs, and in some implementations, head. In one implementation edge detection techniques may be used to extract a position of one or more limbs of the user **116** in the series of images. For example, a main leg angle of a user's **116** leg may be determined, and based on the measurement of this main leg angle over time and from different points-of-view, a three-dimensional model of the leg motion may be generated. The change in position over time of the limbs may be determined and compared with previously stored information to determine an identity of the user **116**, or distinguish one user **116** from another.
(196) In some implementations, identity may be based on a combination of these or other recognition techniques. For example, the user **116** may be identified based on clothing recognition, gait recognition, and facial recognition. The different recognition techniques may be used in different situations, or in succession. For example, clothing recognition and gait recognition may be used at greater distances between the user **116** and the imaging sensors **120**(**1**) or when the user's **116** face is obscured from view of the imaging sensor **120**(**1**). In comparison, as the user **116** approaches the imaging sensor **120**(**1**) and their face is visible, facial recognition may be used. Once identified, such as by way of facial recognition, one or more of gait recognition or clothing recognition may be used to track the user **116** within the facility **102**.
(197) In other implementations other techniques for facial recognition, clothing recognition, gait recognition, and so forth may be employed. For example, facial recognition may use iris detection and recognition, clothing recognition may use embedded RF tags, gait recognition may utilize data from weight sensors **120**(**6**), and so forth.
(198) Block **1008** processes the audio to detect user speech **128** spoken by the first user **116**(**1**) and directed to a second user **116**(**2**) of the facility **102**. For example, the audio data **328**(**2**) may be processed to isolate the sound of the first user's **116**(**1**) voice and to remove background noises such as that of machinery in the facility **102**.
(199) In some implementations, various techniques may be used to determine which user **116** is speaking within the facility **102**. For example, should the facility **102** be busy, multiple users **116** may be speaking simultaneously. These techniques may include TDOA, beamforming, use of data from imaging sensors **120**(**1**), weight sensors **120**(**6**), and so forth. By determining which user **116** is speaking, a speech interaction module **320** may be able to access the user context data **126** associated with that user **116**.
(200) The determination as to whether the first user **116**(**1**) is speaking or not may be based at least in part on one or more of: facial motions of the first user **116** in the acquired images, voice recognition, or sound source localization using the audio from the plurality of microphones **120**(**5**). For example, timestamps of the sensor data **328** may be compared to determine if the image data **328**(**1**) for an image frame in which the user's **116** mouth is open corresponds to a point in time at which audio data **328**(**2**) has been acquired from a microphone **120**(**5**) which is proximate to the position of the user **116** as specified in the position data **336**. By detecting the motion of the user's face **116**, and the presence of corresponding audio, it may be determined that the user **116** is emitting a sound. In another example, voice recognition may be used to distinguish the speech of one user **116** from another. For example, the voice recognition may use pitch, pace, cadence, and so forth to distinguish one user **116** from another. In yet another example, using one or more beamforming techniques, or by sampling data from a plurality of microphones **120**(**5**) some of which are close to the user **116**, some of which are far away, the system may determine that a particular user **116** is speaking.
(201) Block **1010** determines a first user context associated with the first user **116**(**1**). For example, the user context data **126**(**1**) for the identified user **116**(**1**) may be retrieved from the data store **326**. The first user context data **126** may be indicative of one or more of: the historical activity data **334** of the first user **116**(**1**), the gaze direction data **340** of the first user **116**(**1**), position data **336** of the first user **116**(**1**) within the facility **102**, proximity data **338** of the first user **116**(**1**) to the second user **116**(**2**), demographic data **344** of the first user **116**(**1**), user preferences **348** of the first user **116**(**1**), or other information.
(202) In some implementations, the user context data **126** may include information indicative of the identity of other users **116**, such as those who are proximate to the first user **116**(**1**). By knowing to whom the first user **116**(**1**) is speaking, the speech interaction module **320** may be better able to process the user speech **128** as described next.
(203) Furthermore, in some implementations a second user context data **126**(**2**) associated with a second user **116**(**2**) may be determined. The second user **116**(**2**) may be identified as described as above. For example, the second user context data **126**(**2**) for the second user **116**(**2**) may be retrieved from the data store **326**. The second user context data **126**(**2**) may include the identity of the first user **116**(**1**). Stated another way, each of the users **116**(**1**) and **116**(**2**) may have their own respective user context data **126** which is indicative of the identity of the other user **116**.
(204) Block **1012** processes, using the first user context **126**(**1**), the user speech **128** to determine one or more key words or phrases. The key words or phrases may be designated by a system administrator, designated by the user **116**, automatically determined, or a combination thereof. For example, the key words may be the name of an item **104** as stored in the facility **102**, particular adjectives such as “expensive”, particular nouns such as “tote”, and so forth. Continuing the example, as described above with regard to FIG. **9**, the keywords may be “widget”, “cost”, “expensive” and “last time”.
(205) In other implementations where the second user context **126**(**2**) is available, the processing of the user speech **128** may be further based at least in part on the second user context **126**(**2**). For example, by knowing the participants in the conversation, the speech interaction module **320** may be better able to determine actions.
(206) Block **1014** retrieves data responsive to the one or more key words. For example, the speech interaction module **320** may generate a query for cost changes associated with the item **104** named “widget”.
(207) Block **1016** presents output representative of the data with the output device **130**. Continuing the example, the output device **130** may comprise an audio output device **130**(**2**). The output may comprise an audible user interface **132** in which the inventory management module **316** generates user interface data **354** configured to present the audible user interface **132**(**1**), the visual user interface **132**(**2**), or both responsive to the user speech **128**.
(208) As described above, in some implementations the system **100** may be responsive to feedback provided by the user **116**. For example, after receiving the information about the widget, the user **116**(**1**) may find the information of use and respond by saying “thank you”. The second user speech **128**(**2**) as uttered by the first user **116**(**1**) may be deemed indicative of acceptance of the data provided to the first user **116**(**1**). Information indicative of the acceptance may then be stored as historical activity data **334** of the first user **116**(**1**). This information may then be used to modify operation of the speech interaction module **320**, such as by changing weights of one or more elements within a neural network.
(209) In some implementations, the tote **118** may include one or more output devices **130**. For example, the tote **118** may include a display output device **130**(**3**). This output device **130** onboard or coupled to the tote **118** may be used to present a visual user interface **132**(**2**).
(210) FIG. **11** depicts a flow diagram **1100** of another process for performing actions based on user speech **128** between users **116** of the facility **102**, according to some implementations.
(211) Block **1102** determines a user context associated with a first user **116**(**1**) of the facility **102**. As described above, the user context data **126** may

comprise data indicative of one or more of: the historical activity data **334** of the first user **116**(**1**), position data **336** of the first user **116**(**1**) within the facility **102**, proximity data **338** of the first user **116**(**1**), gaze direction data **340** of the first user **116**(**1**) to other users such as a second user **116**(**2**), and so forth.

(212) In some implementations, the user context data **126** may include relationship data **346**. The relationship data **346** may be indicative of a relationship between the first user **116**(**1**) and the second user **116**(**2**). The relationship data **346** may be determined based on one or more of: the first user **116**(**1**) and the second user **116**(**2**) entering the facility **102** together, the first user **116**(**1**) and the second user **116**(**2**) moving together within the facility **102**, or previously stored information indicative of a relationship. For example, facility entry data may comprise log data or other information which indicates the first user **116**(**1**) and the second user **116**(**2**) entered the facility **102** contemporaneously and within a threshold distance of one another. In another example, the user context data **126** may indicate that the first user **116**(**1**) and the second user **116**(**2**) have a relationship because they have been walking proximate to one another for a predetermined length of time within the facility **102**.

(213) The user context data **126** may be received, at least in part, from the user device **134**. For example, the data acquisition module **318** may receive from the user device **134** one or more of information exchanged using a short message service (SMS), information exchanged using an instant messaging service, call metadata associated with an incoming or outgoing telephone or video call, application state data indicative of an application executing on the user device **134**, or other information. For example, the user context data **126** may include text from an instant message "Get more widgets", while the user speech **128** may be the first user **116**(**1**) muttering "well, where are they?" By using the text of the instant message, the speech interaction module **320** is able to process the user speech **128** and determine that the first user **116**(**1**) would benefit from information indicative of the inventory locations **114** of the widgets.

(214) In another example, the call metadata may comprise information such as a telephone number or other network address of the other party(s) with whom communication has been attempted or is established. Other call metadata may include a duration of the call, indication of who originated the call, and so forth.

(215) In yet another example, the application state data may include information such as a browser history, look up history, and so forth. Continuing this example, the first user **116**(**1**) may be looking up technical details about widgets. Based on this information and the user speech **128**, the speech interaction module **320** may provide the information as to the whereabouts of the widgets.

(216) The user context data **126** may include information indicative of a wordbase in use by the user **116**. The wordbase may comprise words, idioms, and phrasing information associated with how the user **116** uses language. Users **116** from different regions or backgrounds may use different wordbases. The speech interaction module **320** may be configured to determine a wordbase of the user **116** based on one or more of word selection of the user **116**, language of the user speech **128**, cadence of the user speech **128**, manual information designating a wordbase of the user **116**, and so forth. For example, the speech interaction module **320** may use demographic data **344** about the user **116**(**1**) to determine that the user **116**(**1**) is from the United Kingdom. As a result, the user speech **128** may be processed using a United Kingdom wordbase, enabling the speech interaction module **320** to correctly determine that the user **116**(**1**) is referring to a flashlight and not a combustible object in the phrase "where is the torch?"

(217) As described above, in some implementations the process may include determining an identity of the first user **116**(**1**). Once identified, user data **342** associated with the identity may be accessed. For example, facial recognition, speech recognition, receipt of an RFID tag, and so forth may be used to associate the first user **116**(**1**) with a particular user account and the corresponding user data **342** associated therewith.

(218) Block **1104** acquires audio of the user speech **128** or other sounds uttered or otherwise generated by the first user **116**(**1**). For example, the microphones **120**(**5**) may generate audio data **328**(**2**). As described above, the user speech **128** may be directed to a second user **116**(**2**). The second user **116**(**2**) may be an identified user of the facility **102**, a guest of the first user **116**(**1**), or a person whom the first user **116**(**1**) is speaking to by way of a two-way radio, telephone, and so forth.

(219) Block **1106** uses the first user context data **126**(**1**) to process the user speech **128**. The processing may include speech recognition, semantic analysis, and so forth. The processing results in a determination of one or more actions which the system **100** may perform to facilitate the user **116**. The actions may include one or more of doing nothing, storing information, retrieving information, presenting information, dispatching assistance from another user **116**, moving inventory **104**, moving inventory locations **114**, and so forth.

(220) Block **1108** performs the one or more actions. For example, the one or more actions may include moving one or more inventory locations **114** from a first position in the facility **102** to a second position in the facility **102**. As described above, the inventory locations **114** may be able to be repositioned within the facility **102**, such as moving from one aisle **112** to another. In another example, the one or more actions may include moving one or more items **104** from a first inventory location **114** to a second inventory location **114**, or from a first position of the facility **102** to a second position. For example, a conveyor, robot, crane, and so forth may be used to move the widget from a high storage rack to a lower level which is accessible to the first user **116**(**1**).

(221) Other actions may include modifying how the first user **116**(**1**) is directed through the facility **102**. A block or module may be configured to access a first proposed route for the first user **116**(**1**) through the facility **102**. For example, the navigation path data **350** data for the first user **116**(**1**) may be retrieved. Another block may then modify the first proposed route to generate a second proposed route. The modification may include adding or removing waypoints, changing a waypoint sequence, and so forth. For example, the modified path may include an addition of a waypoint for the inventory location **114** associated with the item **104** "widget" which the first user **116**(**1**) had mentioned in his user speech **128**.

(222) The actions may further include providing navigational guidance in the facility **102** to the first user **116**(**1**) based on the second proposed route. The navigational guidance may comprise prompts provided by way of an output device which serves to direct the user **116**(**1**) to a particular location in the facility **102**. Continuing the example, the first user **116**(**1**) may be provided with audible prompts stating which direction to turn or visual prompts such as arrows or lines indicating a direction of travel corresponding to the second proposed route. In other implementations navigational guidance may comprise visual projections being projected within the facility **102**, illumination of particular lights within the facility **102**, and so forth.

(223) As described above, the actions may include presenting information by way of the user interface **132**. In some implementations, the user interface **132** and how the user interface **132** is presented may be specific to the user **116**. For example, the user data **342** may include user preferences **348** which specify particular situations within which the visual user interfaces **132**(**2**) are to be used and other situations within which audible user interfaces **132**(**1**) are to be used. Based at least in part on the user data **342**, user interface data **354** may be generated. Also based at least in part on the user data **342**, a particular output device **130** may be selected. Continuing the example above, the first user **116**(**1**) may specify a user preference **348** that when another person is within 2 m of the first user **116**(**1**), the visual interface is to be used and presented on the display output device **130**(**3**) of the user device **134** associated with the first user **116**(**1**). Given that another person is present within the 2 m threshold distance, the user interface data **354** may be configured to provide a visual user interface **132**(**2**). The user interface data **354** may be provided to the user device **134** which has been selected for presentation.

(224) In some situations, the determined action may be modified based on the presence of a third person. For example, the first user **116**(**1**) and the second user **116**(**2**) may be conversing with one another when a third user **116**(**3**) approaches. The block may determine a third person, such as the user **116**(**3**), is within a threshold distance of the first user **116**(**1**). For example, the threshold distance may be 3 m and the third user **116**(**3**) may be 2.5 m away from the first user **116**(**1**), and thus within the threshold distance. User interface data **354** may be generated and may be sent to the output device **130** which is associated with the first user **116**(**1**). The output device **130** may be associated in some implementations by being within the presence of or in the possession of the first user **116**(**1**). For example, the user interface data **354** may be sent to the user device **134** which is being carried by the first user **116**(**1**). In another example, the user interface data **354** may be sent to the tote **118** which is being used by the first user **116**(**1**).

(225) The user speech **128** may comprise an inquiry or may result in an inquiry. The one or more actions may include presentation of information responsive to the inquiry. The information presented in a user interface **132** may be strongly suggestive of the nature of the inquiry, or provide other information about the user context. It may be advantageous to safeguard information about the user context data **126** by obfuscating information presented in the user interface **132**. This obfuscation may include presenting information in the user interface **132** which is responsive to the original inquiry, as well as other information which has been previously inquired about the other users **116**. Thus, the user interface **132** provides the information requested by the user **116**, as well as information which is not a request by the user **116**. To another party observing the user interface **132**, it may not be apparent what information is specifically associated with the user **116** which made the initial inquiry.
(226) One or more blocks execute a first query to retrieve first information associated with the first inquiry. The first query may be responsive to user speech **128** uttered by the first user **116**(**1**). A second query is executed to retrieve second information from a previous inquiry made by a different user **116**, such as from user **116**(**5**). In some implementations the second query may be selected from actions corresponding to user speech **128** previously uttered in other similar circumstances, while in the same area of the facility **102**, and so forth. The user interface data **354** is generated using the first information and the second information.
(227) An output device **130** accessible to the first user **116**(**1**) and other users of the facility **102** is selected. For example, the output device **130** may comprise an overhead speaker or a publicly visible display device **130**(**3**). The user interface data **354** is processed to provide a user interface **132** by way of the selected output device **130**. For example, the user interface **132** may provide the information responsive to the user speech **128** as well as the other information.
(228) Some information which may be presented by the user interface **132** may be deemed sensitive, such that public presentation is discouraged or impermissible. The actions performed by the inventory management module **316** may include one or more of presenting information designated as personally identifiable information, presenting information designated as sensitive, or presenting information about another user **116**, and so forth. For example, the personally identifiable information may include a residential address of the user **116**, information about a customer of the facility **102**, and so forth.
(229) In such a situation, an output device **130** associated with the first user **116**(**1**) may be selected. The association may be determined based at least in part on proximity, user preferences **348**, and so forth. For example, the user preference **348** may specify that sensitive information is to be presented by way of the user device **134** of the first user **116**(**1**), instead of a display device **130**(**3**) which is visible to many users **116**. Once selected, the output device **130** may be used to present the user interface **132**. Continuing the example, the personally identifiable information may be presented on the display device **130**(**3**) of the user device **134**.
(230) The inventory management module **316** may be configured to provide information prospectively which has been previously provided responsive to user speech **128** on other occasions. For example, as the user **116** walks into a particular area of the facility **102** they may be presented with information corresponding to a top three set of inquiries made in that area.
(231) The process for this prospective presentation may include the following. Response data resulting from queries previously generated responsive to speech uttered within a designated area of the facility **102** may be retrieved. For example, response data comprising results of queries responsive to the user speech **128** which has emanated from the users **116** within aisle **112**(**1**) is retrieved from the data store **326**.
(232) The inventory management module **316** may determine the first user **116**(**1**) is present in the designated area. Continuing the example, the position data **336** for the first user **116**(**1**) indicates the first user **116**(**1**) is within the aisle **112**(**1**).
(233) The response data may be presented by way of the user interface **132**. For example, the response data may be used to generate user interface data **354** which is provided to the output devices **130** in the aisle **112**(**1**). The first user **116**(**1**) has now been presented with the information. The user speech **128** uttered by the first user **116**(**1**) after this presentation may seek additional information, resulting in an additional query. This additional query may be executed to retrieve further information. For example, the user speech **128** may indicate that the first user **116**(**1**) would like to have more detail. This information may be retrieved and presented with the corresponding modification of the user interface **132** based at least in part on the information.
(234) The block **1108** may also be configured to perform actions which include requesting permission from the first user **116**(**1**) to take other actions. In this fashion, the speech interaction module **320** may seek confirmation prior to taking other actions. The first user **116**(**1**) may be presented with the user interface **132** which prompts the first user **116**(**1**) for permission to perform an action. For example, the user interface **132** may comprise synthesized speech asking the first user **116**(**1**) "would you like more information on widgets?" Subsequent performance of the action may be responsive to receiving second user speech **128**(**2**) which is indicative of assent to perform the action. Continuing the example, the first user **116**(**1**) may respond with the user speech **128**(**2**) "yes". Following this, the first user **116**(**1**) may be presented with the user interface **132** which includes the additional information on widgets. As described above, in other implementations the user interfaces **132** may be visual, haptic, audible, and so forth.
(235) The actions performed by the inventory management module **316** may also include dispatching a second user **116**(**2**), a tote **118**, robots, or other resource to the first user **116**(**1**). For example, the audio acquired from the first user **116**(**1**) may be processed to determine the presence of non-lexical utterances and item data **332** indicating that a particularly heavy item **104** is being picked, and that the first user **116**(**1**) is having difficulty loading the item **104** onto a tote **118**. The corresponding action taken by the speech interaction module **320** may be to dispatch a second user **116**(**2**) to help with loading.
(236) By using the techniques and systems described in this disclosure, users **116** of the facility **102** may be able to perform their tasks more efficiently and effectively. Instead of having to stop and express a particular inquiry or otherwise directly interact with the inventory management system **122**, the users **116** are able to go about their business within the facility **102** receiving assistance in the form of the various actions which may be undertaken automatically, or after assent by the user **116**.
(237) The processes discussed herein may be implemented in hardware, software, or a combination thereof. In the context of software, the described operations represent computer-executable instructions stored on one or more computer-readable storage media that, when executed by one or more processors, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform particular functions or implement particular abstract data types. Those having ordinary skill in the art will readily recognize that certain steps or operations illustrated in the figures above may be eliminated, combined, or performed in an alternate order. Any steps or operations may be performed serially or in parallel. Furthermore, the order in which the operations are described is not intended to be construed as a limitation.
(238) Embodiments may be provided as a software program or computer program product including a non-transitory computer-readable storage medium having stored thereon instructions (in compressed or uncompressed form) that may be used to program a computer (or other electronic device) to perform the processes or methods described herein. The computer-readable storage medium may be one or more of an electronic storage medium, a magnetic storage medium, an optical storage medium, a quantum storage medium, and so forth. For example, the computer-readable storage media may include, but is not limited to, hard drives, floppy diskettes, optical disks, read-only memories (ROMs), random access memories (RAMs), erasable programmable ROMs (EPROMs), electrically erasable programmable ROMs (EEPROMs), flash memory, magnetic or optical cards, solid-state memory devices, or other types of physical media suitable for storing electronic instructions. Further, embodiments may also be provided as a computer program product including a transitory machine-readable signal (in compressed or uncompressed form). Examples of machine-readable signals, whether modulated using a carrier or unmodulated, include but are not limited to signals that a computer system or machine hosting or running a computer program can be configured to access, including signals transferred by one or more networks. For example, the transitory machine-readable signal may comprise transmission of software by the Internet.
(239) Separate instances of these programs can be executed on or distributed across any number of separate computer systems. Thus, although

certain steps have been described as being performed by certain devices, software programs, processes, or entities, this need not be the case and a variety of alternative implementations will be understood by those having ordinary skill in the art.
(240) Additionally, those having ordinary skill in the art readily recognize that the techniques described above can be utilized in a variety of devices, environments, and situations. Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as exemplary forms of implementing the claims.

Claims

1. A system for presenting information in a materials handling facility (facility), the system comprising: a plurality of microphones configured to continuously acquire audio data representative of sound within the facility; a plurality of imaging sensors configured to acquire images within the facility; an output device; a memory, storing computer-executable instructions; and a hardware processor in communication with the plurality of microphones, the plurality of imaging sensors, the output device, and the memory, wherein the hardware processor is configured to execute the computer-executable instructions to: identify a first user and a second user using the images acquired within the facility; determine a relationship between the first user and the second user, wherein the relationship is determined based on the first user and the second user entering the facility at substantially the same time and within a first threshold distance of one another; process the audio data to detect a portion of the audio data comprising user speech, the user speech comprising natural language spoken by the first user to the second user, wherein the first user and the second user are within a second threshold distance of each other based on the images acquired within the facility; determine a first user context associated with the first user, wherein the first user context is indicative of one or more of: the relationship between the first user and the second user, historical activity of the first user, a gaze direction of the first user, position of the first user within the facility, proximity of the first user to the second user, demographics of the first user, or preferences of the first user; process, using the first user context, the portion of the audio data comprising user speech that is spoken by the first user to the second user to determine one or more key words; retrieve data responsive to the one or more key words; and present output representative of the data with the output device.
2. The system of claim 1, wherein the output device comprises a speaker and the output representative of the data comprises speech sounds, and further comprising computer-executable instructions to: process the audio data to detect second user speech spoken by the first user, wherein the second user speech is indicative of acceptance of the data by the first user; and store information indicative of the acceptance of the data with the historical activity of the first user; and wherein the processing of the portion of the audio data to determine the one or more key words comprises computer-executable instructions to: configure operation of a Hidden Markov Model using the first user context.
3. The system of claim 1, further comprising a tote and wherein: the output device comprises a display device coupled to the tote; the output comprises a user interface presented on the display device; and further comprising computer-executable instructions to: determine the first user is speaking to the second user based on one or more of: facial motions of the first user in the acquired images, voice recognition, or sound source localization using the audio data from the plurality of microphones; and wherein the determination of the first user context is based on the determination the first user is speaking.
4. The system of claim 1, wherein: the identification of the first user and the second user is based at least in part on one or more of: facial recognition comprising: detection of a face in the acquired images, calculation of one or more eigenvectors of the face, and comparison of the one or more eigenvectors with previously stored data; clothing recognition comprising: detection of one or more articles of clothing in the acquired images, classification of the one or more articles of clothing, and comparison of the classifications of the one or more articles of clothing with previously stored data; or gait recognition comprising: determination of one or more user limbs in the acquired images, determination of a change in position over time of the one or more user limbs, and comparison of the change in position over time with previously stored data; and wherein the first user context includes identity of the second user; determine a second user context associated with the second user, wherein the second user context includes the identity of the first user; and wherein the processing of the portion of the audio data comprising the user speech is further based on the second user context.
5. A computer implemented method comprising: accessing sensor data obtained from one or more sensors within a material handling facility (facility); determining a relationship between a first user and a second user, wherein the relationship is based on facility entry data indicative of the first user and the second user entering the facility at substantially the same time and within a first threshold distance of one another; determining, using the sensor data, a user context associated with a first user of the facility, wherein the user context comprises data indicative of the relationship between the first user and the second user; acquiring, using one or more microphones in the facility, audio data of user speech generated by the first user while speaking to the second user of the facility in natural language, wherein the first user and the second user are within a second threshold distance of each other based on the sensor data; using the user context, processing the audio data of the user speech with a computing device to determine one or more actions; and operating, via a network, one or more devices at the facility to perform the one or more actions.
6. The computer implemented method of claim 5, wherein the user context comprises data indicative of one or more items within the facility that the first user is interacting with.
7. The computer implemented method of claim 5, wherein the user context comprises data received from a user device, the data comprising one or more of: information exchanged using a short message service (SMS), information exchanged using an instant messaging service, call metadata associated with an incoming or outgoing telephone or video call, or application state data indicative of an application executing on the user device.
8. The computer implemented method of claim 5, further comprising: determining a wordbase of the first user based on one or more of: word selection in the user speech, language of the user speech, or cadence of the user speech; and wherein the user context further comprises the determined wordbase.
9. The computer implemented method of claim 5, further comprising: determining an identity of the first user; accessing user data associated with the identity; and wherein the user context further comprises the user data.
10. The computer implemented method of claim 9, further comprising: based at least in part on the user data, generating user interface data; based at least in part on the user data, selecting an output device; and wherein the one or more actions comprise: presenting a user interface by way of the selected output device using the user interface data.
11. The computer implemented method of claim 5, wherein the one or more actions comprise one or more of: moving an inventory location from a first position in the facility to a second position in the facility, or moving one or more items from a first inventory location to a second inventory location.
12. The computer implemented method of claim 5, further comprising: accessing a first proposed route for the first user through the facility; modifying the first proposed route to generate a second proposed route; and wherein the one or more actions comprise providing navigational guidance in the facility to the first user based on the second proposed route.
13. The computer implemented method of claim 5, further comprising: determining a third user is within a third threshold distance of the first user; generating user interface data; and sending the user interface data to an output device associated with the first user, wherein the output device is part of a user device or a tote.
14. The computer implemented method of claim 5, wherein the natural language does not comprise an inquiry; and further wherein the one or more actions comprise: generating a first inquiry from the natural language; executing a first query to retrieve first information associated with the first inquiry; executing a second query to retrieve second information from a previous inquiry made by a different user; generating user interface data

using the first information and the second information; selecting an output device accessible to the first user and other users of the facility; and processing the user interface data to provide a user interface by way of the selected output device.
15. The computer implemented method of claim 5, wherein the one or more actions comprising: one or more of: presenting information designated as personally identifiable information, presenting information designated as sensitive, or presenting information about another user; selecting an output device associated with the first user; and presenting a user interface by way of the selected output device.
16. The computer implemented method of claim 5, further comprising: retrieving response data resulting from queries previously generated responsive to speech uttered within a designated area of the facility; determining the first user is present in the designated area; presenting by way of a user interface the response data; and wherein the natural language is processed to generate an inquiry associated with the response data; and further wherein the one or more actions comprise: executing a query to retrieve information associated with the inquiry; and modifying the user interface based on the information.
17. The computer implemented method of claim 5, wherein the one or more actions comprise: prompting the first user for permission to perform an action; and wherein the performing the action is responsive to receiving second user speech indicative of assent to perform the action.
18. A computer implemented method comprising: acquiring, using one or more microphones connected to a network at a material handling facility (facility), audio data representative of sound at the facility; acquiring, using one or more sensors at the facility that are connected to the network, sensor data; determining, with one or more computing devices, a relationship between a first user and a second user, wherein the relationship is determined based on the first user and the second user entering the facility at substantially the same time and within a threshold distance of one another; processing the sensor data using the one or more computing devices to determine a user context of the first user at the facility, wherein the user context is indicative of the relationship between the first user and the second user; recognizing, with the one or more computing devices, a sound uttered by the first user and directed to the second user at the facility in the acquired audio data; determining, with the one or more computing devices, an action based on the user context and the sound that is recognized; and sending from the one or more computing devices, via a network, commands to perform one or more actions associated with operation of the facility.
19. The computer implemented method of claim 18, wherein the sound is a non-lexical utterance.
20. The computer implemented method of claim 18, wherein the one or more actions comprise dispatching a third user to assist the first user.
21. The computer implemented method of claim 18, wherein the one or more actions comprise: retrieving information; presenting a first user interface comprising a request for permission to present the retrieved information; receiving input from the user interface indicative of approval to present the retrieved information; and presenting a second user interface comprising at least a portion of the information.